

GE Healthcare

LightSpeed 7.X Installation Manual

(Book 2 of 2)

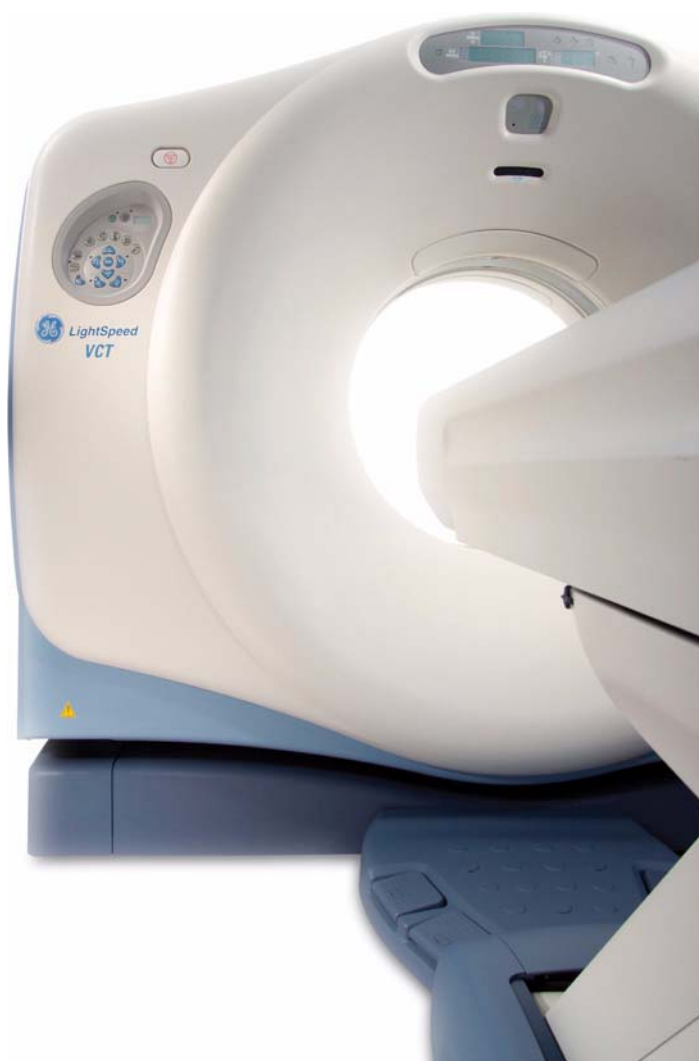
LightSpeed VCT XT

LightSpeed VCT

LightSpeed VCT Select

LightSpeed Pro32

LightSpeed VCT Limited Access



OPERATING DOCUMENTATION



5116411-100
Rev 15

Book 2 of 2: Electrical Calibration, Integration & Testing

Pages 153 - 286

Effectivity

The information in this manual applies to the following LightSpeed 7.X CT Systems:

- LightSpeed VCT (64-Slice)
- LightSpeed VCT Select (32-Slice)
- LightSpeed Pro32 (32-Slice)
- LightSpeed VCT Limited Access

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Chapter 4

Electrical Introduction

Section 1.0 Installer/FE Notices

1.1 Shipping, Warehouse and Transportation Warning

- This gantry is designed to be moved using the shipping dollies and should not be lifted or moved using a lift truck.
- Do Not Hoist Gantry or Table using Dollies.

1.2 International Shipments

- Dollies must be used to remove the gantry from the shipping skid and to transport the gantry to the customer's site.
- If lifting is required, refer to the Pre-Installation Manual for instructions.

1.3 On Site Warning

This system requires a gantry bearing gap inspection *before* electrical calibration is started. See [Gantry Bearing Gap Inspection, on page 43](#).

1.4 Service Actions

Open a dispatch and record the bearing inspection results first, then close the dispatch and continue with the electrical calibration procedures.

Purpose:



CAUTION

Shock Hazard.



Voltage Present.

No service on left side while energized.



NOTICE

**Potential for
Data Loss and/
or Equipment
Damage**

To prevent potential data loss and equipment damage, please do the following:

- **Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 6](#) of this book.**
- **Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.**

Section 2.0

Introduction

Use the continuity and ground checks to verify the system power connections have not shorted to ground and that the ground and neutral connections are intact.

Section 3.0

Review Mechanical Hand Off Material

Complete the Mechanical Hand Off checklist:

- ☐ All options were installed. If not, contact your install specialist.
- ☐ Check for short ships.
- ☐ Complete paperwork and phone calls as needed.
- ☐ Review cable connections with mechanical team.

Section 4.0

VCT Calibration Training Requirements

See requirements outlined in Book 1.

Section 5.0

Required FE Common Tools and Supplies

Note: Items with “checks” (✓) are included in the Install Support Kit. Special CT tools used for mechanical installation and service are included.

5.1 FE Calibration and Service Tool List

- Standard FE Tool Kit – See new VCT tool list.
- Fluke 87 DVM or equivalent
- AC Clamp on amp meter
- Dale 600 or 601
- VCT QA Phantom

5.2 Electrical Tools

These tools must be calibrated yearly.

- Fluke 87 DVM or equivalent
- Clamp-on amp meter
- Dale 600 meter

5.3 Image Quality Calibration Tools

- VCT QA Phantom
 - 35 cm Poly
 - Phantom Holder

5.4 Optional Tools

Additional tools may be required if warranty recalibrations are required.

5.5 Safety Materials

- Note: Items with “checks” (√) are included in the Install Support Kit
- √ Lockout/Tagout kit, or equivalent
 - Safety Glasses

5.6 Cleanliness

The VCT system must not be located in, near or around construction. The room should be clean and clear of construction dust and installation materials.



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.
DUST COVERS REQUIRED for installations on construction sites.

Section 6.0 Requirements/Assumptions

- This procedure will be performed by an appropriately trained GE engineer.
- You need the Internet (IP) addresses the first time you execute a reconfig on the system.
 - When you connect the system to a network, contact the system administrator to obtain the IP addresses for all the computers in the suite.

Section 7.0

FE Workflow

- 1.) Review mechanical hand-off material.
- 2.) Obtain required FE common tools and supplies.
- 3.) Perform electrical power-on and ground checks.
- 4.) Gather all customer information needed for reconfiguration.
- 5.) Perform computer integration.
- 6.) Complete Table/Gantry integration.
- 7.) Complete the calibration process.
- 8.) Perform tube warm-up and fast calibration.
- 9.) Complete tomographic plane indication.
- 10.) Run image series tests.
- 11.) Run system functional test.
- 12.) Create system state DVD.
- 13.) Perform Patient Touch Leakage test.
- 14.) Perform the CT System Chassis Leakage test, as required by local code.
- 15.) Complete installation and verification of any customer options.
- 16.) Complete and return GE Form e-4879 Installation Data Verification for all installations.

Section 8.0

Checklists for Completed Installation

8.1 System-Level Tasks

Complete the tasks listed and check the appropriate box on the GE e-4879 form (explained in [Section 8.0 on page 177](#)).

8.1.1 General

- ☐ HVAC system is operational and environmental data reported on the GE e-4879 form.
- ☐ System realignments completed, if required.
- ☐ Broadband installed and operational.
- ☐ Power and ground audit completed.

8.1.2 Optional and Regional

- ☐ Seismic mounting kit installed, if required in your area.
- ☐ Generator recalibration completed, if necessary.
- ☐ Collimator recalibration completed, if necessary.

8.2 Site Clean-Up

- ☐ All DVDs for customer options placed in the GE service cabinet.
- ☐ All system software and service tools placed in the GE service cabinet.
- ☐ System cleaned and nicks touched up with paint.
- ☐ Room is cleaned and all trash disposed of properly.
- ☐ Installation site cleaned and all trash properly disposed.

8.3 Dolly Return

- ☐ Return of dollies arranged and dolly pick-up confirmed.

8.4 Options

Check the appropriate box on the GE e-4879 form to verify the installation and proper functionality of all customer-ordered options.

- ☐ Injector installed and operational.
- ☐ Advantage Windows Workstation installed and functional tests completed.
- ☐ Advantage 4D installed and functional tests completed.
- ☐ Filming/Camera/DASM installed and operational.
- ☐ Modem installed and functional tests completed.
- ☐ UPS installed and functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Customer software options installed and operational.
- ☐ Teleradiology connections completed. (See [Section 10.0 on page 257.](#))
- ☐ Remote monitor installed and operational.
- ☐ Bar Code Reader installed and operational.
- ☐ Cardiac monitor and stand installed and operational.

8.5 Final Activities (Paperwork)

- ☐ GE e-4879 completed; see [Section 9.0](#). (Required for installations in ALL countries.)
- ☐ FDA 2579 completed; see [Section 9.0](#). (Required ONLY for U.S. installations.)
- ☐ Any PQRs or PSRs encountered have been reported.
- ☐ All FMIs for system completed, if necessary.
- ☐ All dispatching activities (03-04-10 codes) completed.
- ☐ Customer acceptance checks completed.
- ☐ System transfer completed and appropriate GE Healthcare personnel notified.
- ☐ All outstanding customer installation issues have been addressed.

Section 9.0

GE and Regulatory Forms

Field Engineers must complete and submit the documents listed in Section 8.1 for ALL installations, regardless of the country. In addition, for installations performed within the United States, Field Engineers must ALSO complete and submit the documents listed in Section 8.2.

9.1 All Countries

9.1.1 GE e-4879 Form

The Field Engineer should:

- 1.) Locate the GE e-4879 form on the Service CD.
- 2.) Complete the form.
- 3.) E-mail the completed form to the following:
 - a.) HHS Administrator
 - b.) Babar.maqsood@med.ge.com (Engineering/Product Development)

9.1.2 Product Locator Cards

The Field Engineer should:

- 1.) Enter the Product Locator Card information on the Product locator Website. Go to the following address to access the site: http://gib.gehealthcare.com/gib/gib_entry.jsp
- 2.) Leave one (1) Product Locator Card (or a copy) at the customer site for EACH piece of equipment installed there.

Note: CT Manufacturing completes the GE HHS Data Sheets and provides them to the HHS Administrator.

9.2 U.S. Installations Only

9.2.1 FDA 2579 Form

The Field Engineer should:

- 1.) Download the FDA 2579 form from the HHS Support Central Web site: http://supportcentral.ge.com/products/sup_products.asp?prod_id=16442
- 2.) Complete the form.
- 3.) E-mail the completed form to the HHS Administrator.

Note: Do NOT print this form after completion. The HHS Administrator will e-mail a printable version to the FE for customer site records.

Note: **Some states require a State Registration Number to complete this form. For any questions concerning your state, contact the HHS Administrator or check the HHS Support Central Website.**

Some states may also require additional information and test information. For instructions, contact the Project Manager of Installation.

9.2.2 System Chassis Ground Leakage Test Form

The Field Engineer should:

- 1.) Locate the System Chassis Ground Leakage Test form on the Service CD.
- 2.) Complete the form, if it is required in the local area.
- 3.) Forward the results as instructed on the form.

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Chapter 5

Electrical Integration and Safety Verifications



CAUTION



Shock Hazard.
Voltage Present.
No service on left side while energized.



NOTICE

**Potential for
Data Loss and/
or Equipment
Damage**

To prevent potential data loss, please do the following:

- When directed, record data collected from procedures in this chapter into Form F4879, located in [Chapter 6](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Electrical Power On & Ground Checks



WARNING



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES.
USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

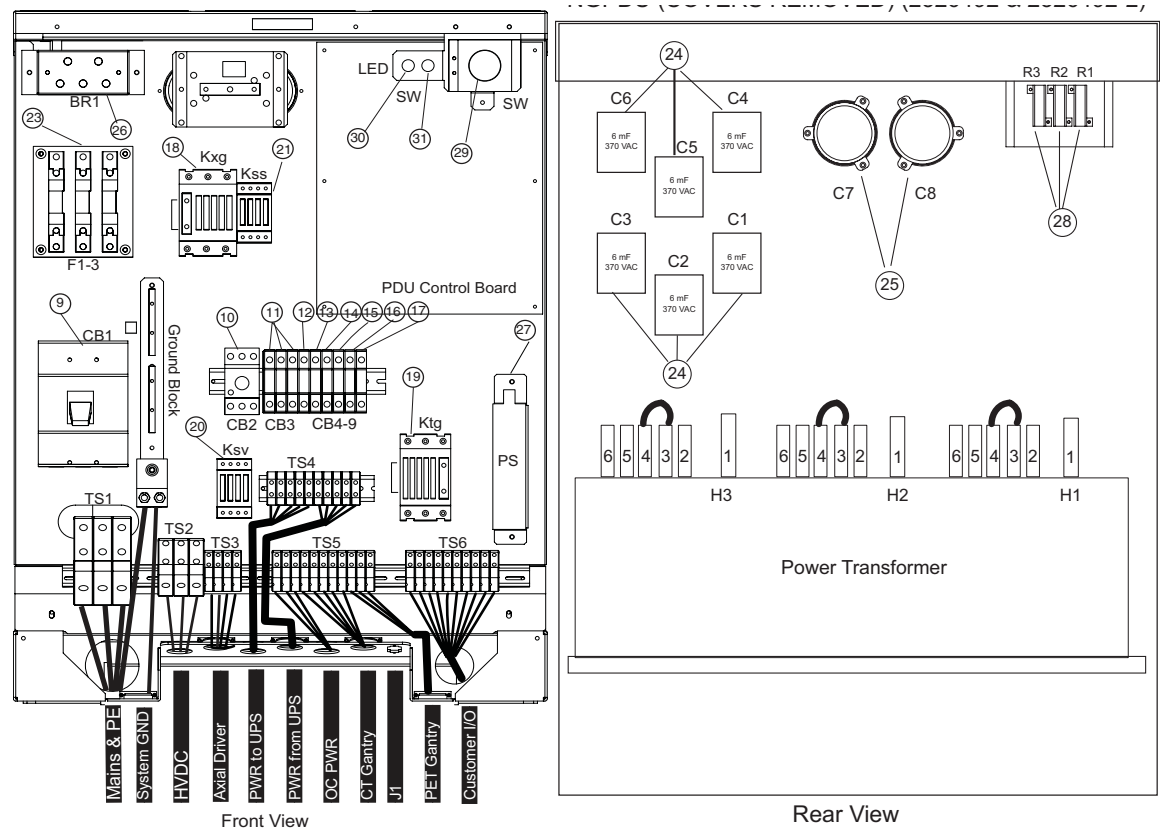
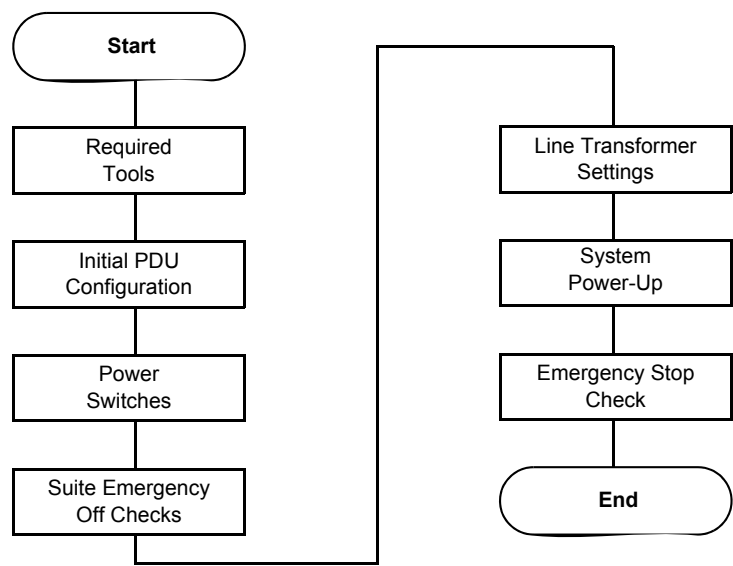


Figure 5-1 NGPDU

1.1 Introduction and Flowchart



1.2 Electrical Power On & Ground Checks Process Overview

1.3 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

1.4 Initial PDU Configuration



WARNING



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

1.4.1 Circuit Breakers

Set all PDU, gantry, console, and table circuit breakers to OFF.

1.4.2 Relay Board

- 1.) Set SW 2 to the Auto-Off position.
- 2.) When the system is powered, three lamps should be "ON". (Refer to [Figure 5-2.](#))

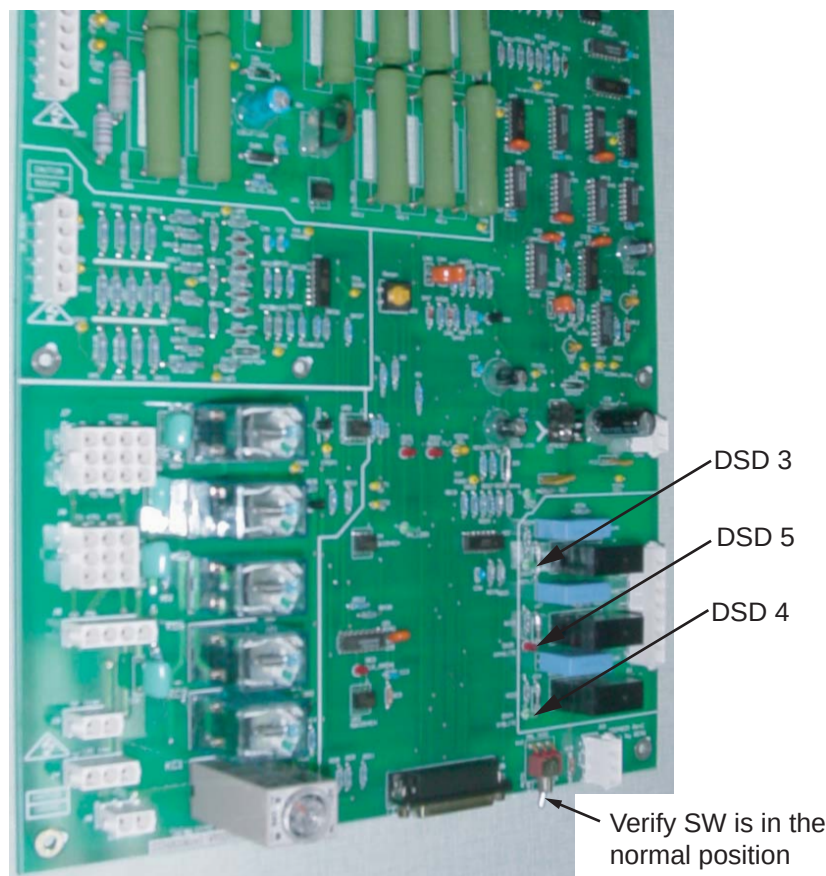


Figure 5-2 NGPDU Control Board

1.4.3 Power Switches

Turn OFF all system power switches at their subsystems.

- Gantry power pan breaker
- Gantry slip ring ground fault breaker on stationary base right side & all service switches
- Table base power
- Console power

1.4.4 Hardware and Connection Check

Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

- PDU
- Gantry
- Table
- Console

1.4.5 Covers

Install, or verify the presence of, all the lexan safety covers for the PDU.

1.5 Suite Emergency Off Checks SEO



WARNING

VERIFY ALL PERSONNEL HAVE CLEARED THE SYSTEM BEFORE YOU TURN ON WALL POWER.

- 1.) Turn wall power ON to the PDU.
- 2.) **Press the suite emergency off button and verify it turns off wall power to the PDU.**
(Typically, this red palm button is located on the wall close to the console, within the scan suite.)
- 3.) Verify that all "Emergency Off" buttons are working properly.
- 4.) Leave power "OFF".

1.6 Line Transformer Settings

1.6.1 Requirements

- 1.) The PDU is shipped configured for 480VAC.
- 2.) Complete only if your site uses a voltage other than 480VAC.
- 3.) If PDU is configured for 480VAC, go to [1.7](#). Otherwise, proceed to [Section 1.6.2](#).



WARNING

MAKE SURE YOU TURNED OFF, TAGGED AND LOCKED THE MAIN WALL POWER BEFORE YOU CHANGE TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION. TURN OFF WALL POWER TO CONNECT OR MOVE METER LEADS, OR TO REMOVE OR INSTALL COVERS.

1.6.2 Line Input Conditions

- 1.) Monitor the No Load Line to Line Voltage at L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 2.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 5-1](#) and [Figure 5-3](#) for tap locations).

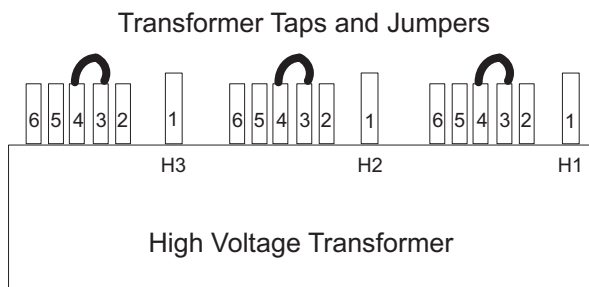


Figure 5-3 PDU Tap Positions (Rear)

Note: Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.

- 3.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum and maximum values listed in [Table 5-1](#).
- 4.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at L1, L2, & L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

NO LOAD Line to Line Voltages		TAP CONNECTIONS (All 3 phases must have same the configuration)		
Nominal	Maximum Range (10%)	Phase A Connection	Phase B Connection	Phase C Connection
480V*	432 to 528*	3-4*	3-4*	3-4*
460V	414 to 506	3-5	3-5	3-5
440V	396 to 484	3-6	3-6	3-6
420V	378 to 462	2-4	2-4	2-4
400V	360 to 440	2-5	2-5	2-5
380V	342 to 418	2-6	2-6	2-6
240V**	216 to 264**	1-4**	1-4**	1-4**
220V**	198 to 242**	1-5**	1-5**	1-5**
200V**	180 to 220**	1-6**	1-6**	1-6**
* Factory Default				
** 2326492-3 PDU only				

Table 5-1 PDU Line Tap Connections

Record system voltages here:

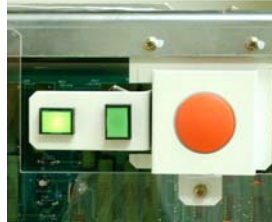
Phase A: _____ Phase B: _____ Phase C: _____

1.7 System Power-Up

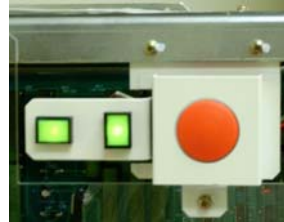


CAUTION Verify all personnel have cleared the system before you turn on wall power.

- 1.) Turn ON the A1 breaker panel.
- 2.) Turn ON all system power switches and breakers (PDU, gantry, table, console).
 - All PDU breakers
 - Make sure that the on/off button (on the front PDU panel) is ON for console power.



PDU Power Switch Off



PDU Power Switch On

- Gantry power pan breaker
- Gantry slip ring ground fault breaker on stationary base right side & all service switches
- Table base power
- Console power (Check internal breaker.)

SUB-SYSTEM POWER-UP

- 1.) Turn ON switch S3 in the table (120VAC 24-hr power).
- 2.) Turn the gantry **120 - 208VAC** to ON. (Light should turn on.)
- 3.) Turn **AXIAL DRIVE ENABLE** ON. (Light should turn on.)
- 4.) Turn **HV DC ENABLE** ON. (Light should turn on.)
- 5.) Push the Service Switch Panel reset button. (See [Figure 5-4](#).)

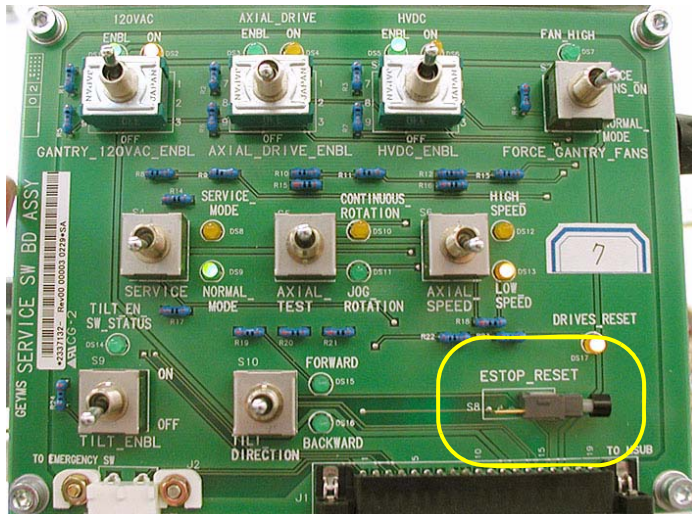


Figure 5-4 Service Switch Panel

AXIAL ENABLE SWITCH TEST

- 1.) Unplug all top cover fan plugs.
- 2.) Turn OFF axial drive enable switch AXIAL_DRIVE on the Service Switch Panel.
- 3.) Clear the gantry area for rotation.
- 4.) Press the alignment light push button.
- 5.) Verify that the gantry did not rotate.
- 6.) Turn ON axial drive enable switch AXIAL_DRIVE on the Service Switch Panel.

Note: For the initial condition, do NOT leave the tube at the 2:30 position.

ROTATION SAFETY CHECKLIST

- 1.) Turn OFF axial drive enable switch AXIAL_DRIVE.
- 2.) Turn OFF HVDC enable switch.
- 3.) Press red E-STOP button.
- 4.) Manually rotate the gantry 360 degrees.
 - Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - Listen for any loose parts.
(Tighten parts as needed.)
- 5.) Turn ON all enable switches.



WARNING

MAKE SURE THERE ARE NO OBSTRUCTIONS AROUND THE GANTRY. PRESSING THE ALIGNMENT LIGHT PUSHBUTTON WILL CAUSE THE GANTRY TO ROTATE.

- 6.) Press the alignment light push button.
- 7.) Verify that the gantry rotates.
- 8.) Turn off the laser light.
- 9.) Perform a 2-second X-ray OFF scan.



NOTICE

During the scan, it may be necessary to enter the scan room to obtain a better listening position. If so, keep a finger on one of the four E-STOP buttons (on the gantry), to quickly stop the gantry, if necessary.

- a.) From the console, click on the SERVICE DESKTOP icon.
- b.) Select DIAGNOSTICS.
- c.) Select DIAGNOSTIC DATA COLLECTION
- d.) Set the scan time to 4.00 seconds and rotating X-ray Off.
- e.) Select ACCEPT.
- f.) Leave the door open. (This makes it easier to hear any loose or interfering parts.) The gantry should spin for approximately 45 seconds
 - * Listen for any interference between the rotating and stationary parts.
(Correct any interference problems.)
 - * Listen for any loose parts.
(Tighten parts as needed.)
- g.) After completing the 4-second scan, repeat Step a through Step f, with the following scan times:
 - * 2.0 second scans
 - * 1.0 second scans
 - * 0.7 second scans
 - * 0.5 second scans

1.8 Emergency Stop Check

- 1.) Use the gantry push-buttons to advance the cradle about 0.5m (2ft) from the home position.
- 2.) Press one of the E-STOP buttons on the gantry.
- 3.) Make sure the TABLE POWER shuts off, and the green LED flashes.
- 4.) Depress one of the table elevation buttons, to verify the emergency stop disabled table elevation.
- 5.) Depress one of the cradle drive buttons, to verify the emergency stop disabled the cradle drive.
- 6.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER. (120 VAC LED stops flashing.)
- 7.) Press the other E-STOP button on the gantry.
 - a.) Make sure the Table Power shuts off.
 - b.) Manually move the cradle to the home position to make sure the cradle clutch released.
 - c.) Make sure the cradle latches securely in the home position.
- 8.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER.
- 9.) Press one of the four table tape switches to make sure the table down motion stops. Repeat with the three remaining table tape switches.
- 10.) Press the console emergency stop switch; make sure the Table Power shuts off.
- 11.) Press one of the **RESET** buttons to turn on X-RAY DRIVES POWER. (See [Figure 5-5](#)).



Figure 5-5 Reset buttons on Gantry and Service Switch bank.

Note: Emergency Stop buttons are located on the front and rear of the gantry (8 in all), as noted in [Figure 5-6](#). They are also located on both sides of the table base (4 in all). Additionally, an emergency stop button is provided on the Operator Console SCIM (see [Figure 5-7](#)).



Figure 5-6 Gantry Emergency Stop Button Positions

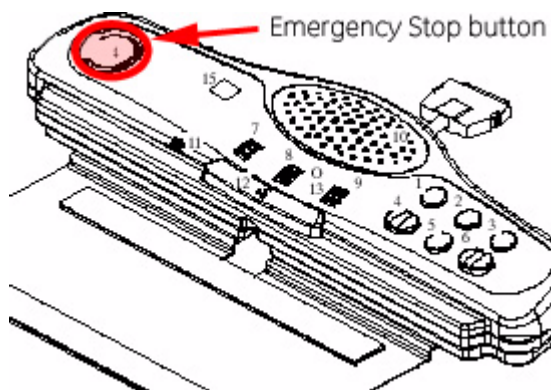


Figure 5-7 SCIM Emergency Stop Button

Section 2.0

Computer Integration and Configuration

2.1 Introduction and Flowchart

This section describes the reconfiguration, system state restore, options installation, and monitor adjustment procedures.

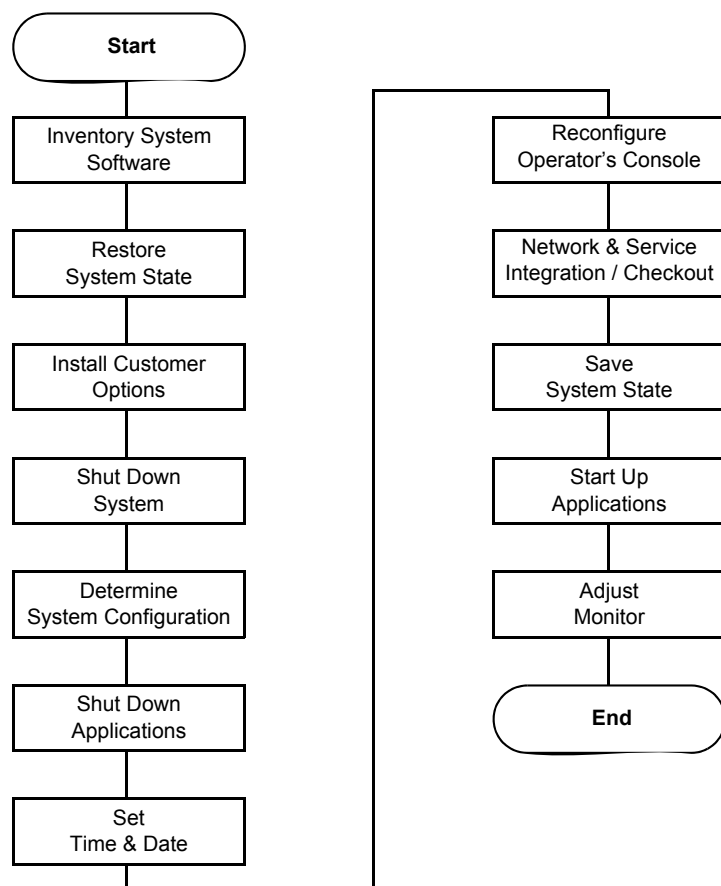


Figure 5-8 Computer Integration and Configuration Process Overview

2.2 Inventory System Software

Locate the box with the system software and option DVD disks, system order sheets, product locator cards, system reconfig DVD and HHS paperwork.

You should find the following software CD documents:

- System Operating Software CD set.
- VCT Learning and Reference Guide
- VCT Tip Simulator
- CT Advanced Applications
- LS 7.x VCT & Pro 32 Service Information

Note: There may be other items in addition to those above.

2.3 Restore System State

Tools Required: None

Your system should have a system state DVD, located in the software collector box.

The system state DVD contains:

- Characterization
- Calibrations
- Gen Cal
- Other Data

If you cannot locate the shipped system state DVD, you must do a **complete recalibration** of your system.

The installation process uses all the system state files. At this time, use the system state DVD to restore all files.

- 1.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.
- 2.) Click the UTILITIES icon.
- 3.) Select SYSTEM STATE.
- 4.) Insert the DVD into the DVD drive.
- 5.) Select CHARACTERIZATION and CAL.
- 6.) Select CALS.

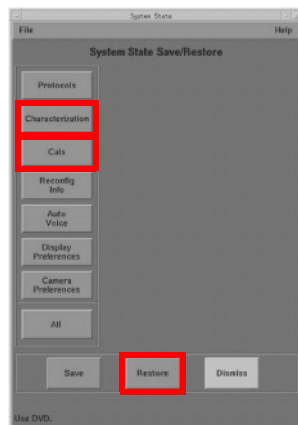


Figure 5-9 System State Restore

- 7.) Select RESTORE to restore the system characterization and phantom calibration files to the system.

Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.

If any error should occur during the restore process, see the Software Load Procedure manual (Load From Cold) for information regarding possible error messages and their recovery.

- 8.) Click NO for the Reset Scan Hardware popup message.
9.) Click DISMISS.

2.4 Install Customer Options

Tools Required: Laptop

2.4.1 System Options

Note: Your system has one or more DVDs that contain customer-purchased options. Standard options are required for system operation. Install the Options DVD(s) at this time, following the instructions in this section. A more complete listing of options can be found on the Service CD-ROM shipped with your system.

The following tables are for clarification and identification of options for installation purposes only.

Note: Standard options are loaded before additional options. Load the standard options in the order listed in Table 5-2.

Catalog Number	Part Number	Marketing Name	Installed Options	Comments
B7530LP	2371127	Smart Prep	SmartPrep	
B7868ML	2360576	Auto MA	AutomA	
B7868ES	2370921	Large Image Series	3000 Image Series	
B7500LN	2371407	Connect PRO	Connect Pro	Required for Exam Split
B7864KH	5133030	Exam Split	Exam Split	- Requires Connect Pro - Before installing, engineer needs to determine if Hard Exam Split (HES) or Virtual Exam Split (VES). - VES requires PPS (Performed Procedure Step), which is installed as a part of the Connect Pro install.
B7810LS	2360525	Direct 3D	Direct-3D	
B7864KG	5133028	DMPR	Direct-MPR	Direct-MPR requires 4 GB of Host Main memory as shown in hinv; otherwise, with 2 GB the maximum images is set to 1200.
B 7864KJ	5133032	Data Export	Data Export	
B7864KK	5133034	Interchange, CD/DVD	Copy Composer	

Table 5-2 Standard Options

Additional options must be loaded in the following order for proper operation.

Marketing Name	Installed Options	Comments
Neuro 3D Filter	NeuroFilter	
AutoFilter-and-Transfer	AutoFilter-and-Transfer	
VCT option (64 slice)	Patient-64-Slice	
VCT-Hi-Power Option	VCT-Hi-Power Option	
Smart Score Pro	SmartScore Pro	Requires AutomA for EKG Monitor Required for Cardiac options
CardIQ Snapshot	CardIQ SnapShot	
.35 Second Rotation	Sub-0.4-Second-Scan	Requires CardIQ SnapShot
ECG Trace OC	EKG Viewer	Requires CardIQ SnapShot
Cardiac Enhancement Filter	Noise Reduction Filter	Requires CardIQ SnapShot
SnapShot Pulse	CardIQSnapShot-Cine	
DentaScan	DentaScan	
CTPerfusion3Neuro	CTPerfusion3Neuro	
Volume Viewer	VolumeViewer	Install before DMPR
CardIQPlus OR Cardiac Pro on OC	CardIQXpressPlus OR CardIQ2XpressPro	Requires CardIQ SnapShot
Card EP	CardEP	
AdvVesselAnalysis	AdvVesselAnalysis	
Auto Bone	AutoBone	
CTCBase	CTColonoBase	load only one
CTCPlus	CTColonoPlus	
CTC PRO on OC	CTColonoPro	
SmartStep	SmartStep	
Volume Shuttle	AxialShuttle	

Table 5-3 Additional Options

- 1.) Locate all of the option DVDs.
- 2.) Ensure that the Options DVD(s) are NOT write-protected at this time. The initial install requires that the DVD be write-enabled; subsequent installs can be done with the DVD write-protected.
- 3.) Inventory product locator cards. There should be approximately 28 cards depending on the options ordered. Use the sales order sheet that comes with the system to verify that all sold items are present.
- 4.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon.
- 5.) Click the CONFIGURATION icon.
- 6.) Click INSTALL OPTIONS.
A blank Options window is displayed (Figure 5-10).

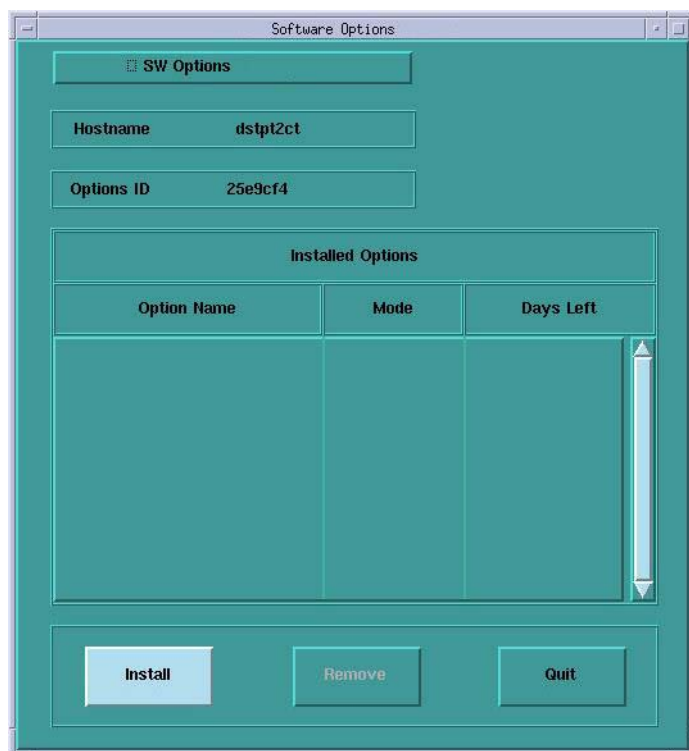


Figure 5-10 Install Options Window

7. Click INSTALL. A **Select Mechanism** (Figure 5-11) prompt appears. Click the button for the mechanism through which Option Keys will be installed.



Figure 5-11 Select Mechanism window

- 8.) Click PERMANENT. A **Select Device** window (Figure 5-12) opens.

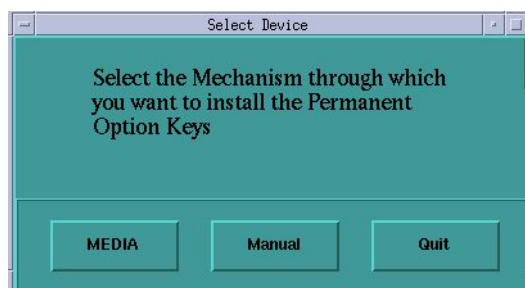


Figure 5-12 Options Window when First Selected

9. Click the MEDIA button.

- 10.) Insert the options DVD into the DVD drive and click on OK. (If you do not have an options DVD, click on OK anyway, wait for the abort pop up, then abort the process.) Available options on the DVD appear on the **Software Options** window. (See [Figure 5-13](#).)

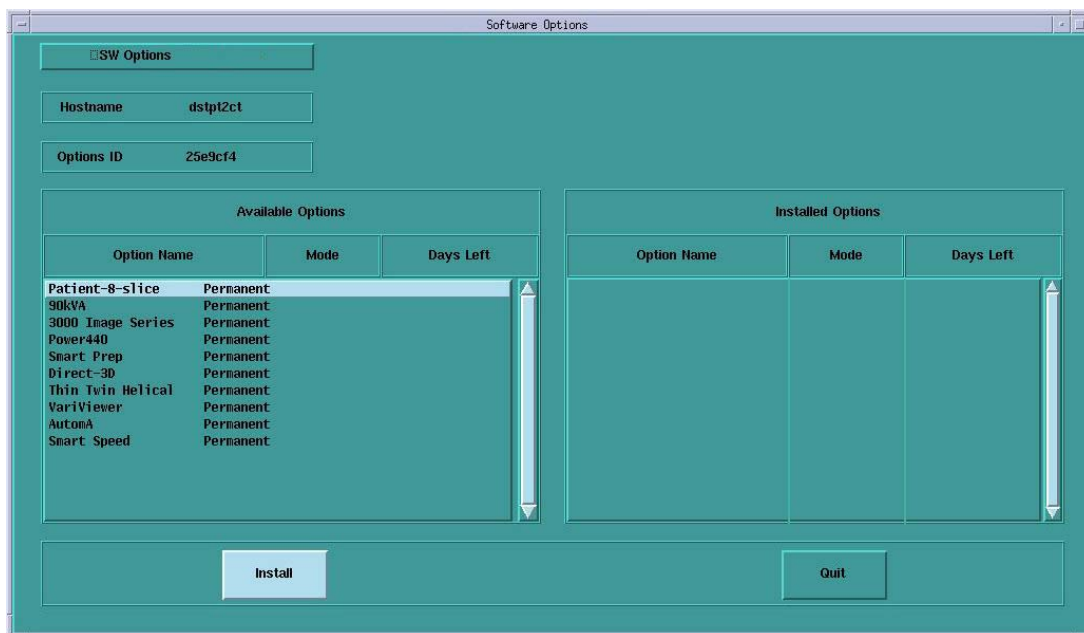


Figure 5-13 Software Options screen

- 11.) Select each option in the left-hand column in the order given in [Table 5-2](#) to install the corresponding software. The options must be loaded in the order given in [Table 5-2](#).
- Note: Some options require that data field(s) be completed. Refer to the inventory sheet for the options with data requirements. Have this information available when completing this section.
- Check the FDO to see what options were ordered.
 - Compare FDO options to those on the Options DVD.
 - If different, contact your local sales representative.
- 12.) Select INSTALL. A box may appear while the options are loading. When an option is displayed in the *Installed Options* list on the right side, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half minute.
- 13.) When all options have been installed, check the permanent options on the right side of the screen against those ordered. Make changes as required and close this screen.
- 14.) Run Verify Options to confirm that all options are loaded.
- 15.) After the options are installed, select QUIT then QUIT again.
- 16.) Remove the DVD and write-protect the side with options.
- 17.) When the system prompts to reboot, click YES, and reboot the system to complete the installation.

2.4.2 Camera

Tools Required:

- Small flat blade screw driver
- Data collected from data sheets (See “Camera Application Configuration” on page 275.)
- Software Load Procedures manual
- System Service manual.

Note: If a DASM is required, the DASM hardware must be installed before proceeding.
For details on camera configuration, refer to the Software Load Procedures manual.
For details on troubleshooting the camera, refer to the System Service manual.

- 1.) Click on the SERVICE DESKTOP icon.
- 2.) Select UTILITIES icon.
- 3.) Select INSTALL CAMERA.
- 4.) Select ADD.
- 5.) Select DASM or DICOM.
- 6.) Follow procedures on the screen.
- 7.) Return to Home Page

2.5 Shut Down System

- 1.) Click the SERVICE DESKTOP icon.
- 2.) Click SHUTDOWN
- 3.) Click REBOOT.
- 4.) Restart the system.

2.6 Determine System Configuration

For convenient removal and use during installation, System Configuration Data Sheets appear in [Appendix E on page 273](#).

Below is a summary of key information, some of which is required from the customer, to complete system configuration. When gathering this information, refer to [Reconfigure Operator Console \(OC\) on page 198](#).

System File Information:

- Hospital name (Ask the customer for ALL related fields.)
- Service ID
- Patient Info – Next MOD#, Exam #, Diagnostic # 50000 default.
- Click YES to regenerate database.
- Click NO for Mobile System.

Preference File Information

- Doctors title
- Date Format
- Time Format
- Language type
- Selected Fast Cal KV's – default - ALL unless instructed otherwise.
- Dose Information – default - Unless instructed otherwise.
- Dicom – default - Unless instructed otherwise.

Hardware File Information

- Select table type GT 2000 or GT 1700
- Default for all others
- Network printer – default

Network file Information

- Suite Name – (from FE or hospital)
- Host Name – (from FE or hospital)
- IP Address
- Net Mask
- Broadcast Address
- Default Gateway
- Advanced options – default - Unless instructed otherwise by the FE.

2.7 Shut Down Applications

2.7.1 Standard Level

If Applications is currently running, you must shutdown system applications.

- 1.) Click on the SERVICE DESKTOP icon.
- 2.) On the desktop toolbar, select UTILITIES icon.
- 3.) Click APPLICATIONS SHUTDOWN (to bring down applications only).

2.7.2 Super User Level

- 1.) Open a Shell window.
- 2.) Type: `su` - ENTER
- 3.) Type the root (super user) password and press ENTER

2.8 Enable the Console Laptop Port

On the console in a unix shell change user to root and enable the console laptop port:

- 1.) Type: `su` - ENTER
- 2.) Type the root password and press ENTER
- 3.) Type: `enableFEport` ENTER

2.9 Reconfigure Operator Console (OC)

Note: The document collector box that arrived with your system contains the *Software Installation Procedures* manual, which documents the reconfiguration procedure in more detail.

2.9.1 Overview

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a Shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

```
/var/adm/install.log.YYYYMMDDWWWHHMMSS
```

Where

YYYYMMDDWWWHHMMSS is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

2.9.2 Procedure

- 1.) Shut down applications from the Service Desktop.
- 2.) In an **xterm** window, log in as root:
 - a.) Type: `su -` ENTER
 - b.) Type the root password and press ENTER
- 3.) Launch the Install utility:

Type: `reconfig` ENTER at the prompt.

The OC displays the Install Utility Window as shown in [Figure 5-14](#).

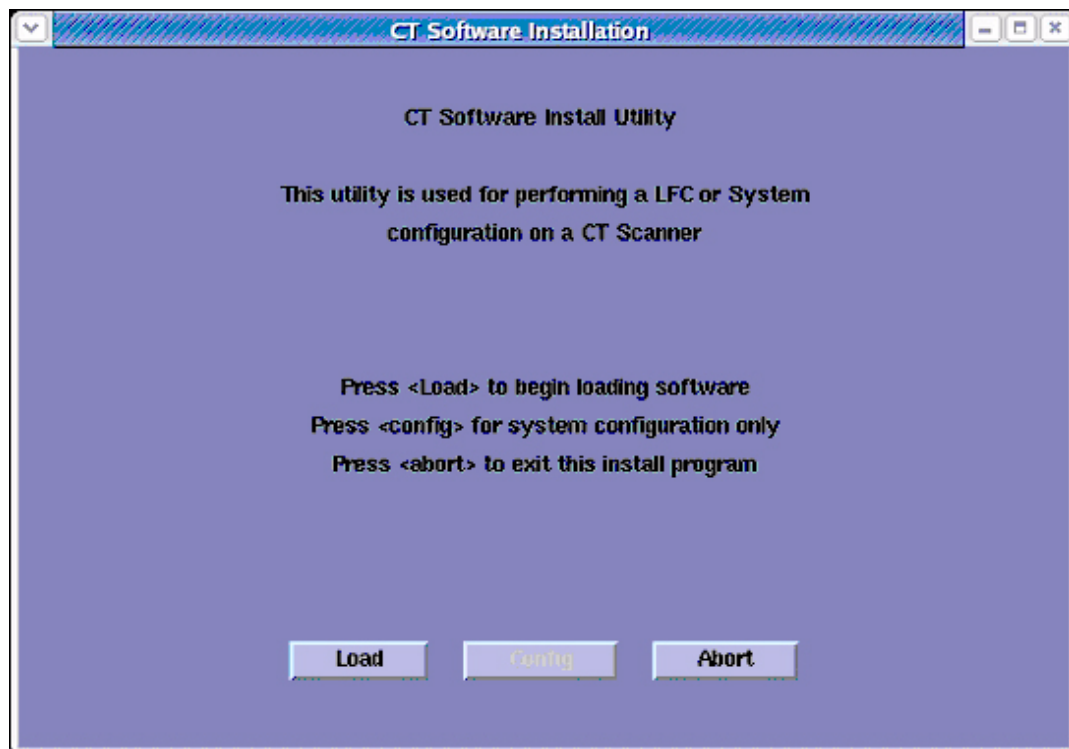


Figure 5-14 Install Utility Window

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

4.) Click the CONFIG button.

The OC displays the System Configuration - System Settings Screen as shown in [Figure 5-15](#).

Figure 5-15 System Settings Screen

5.) Configure System Settings

- a.) Hospital Name (Figure 5-15, item a) configures the name that will show up on images produced by this scanner. *Example:* ST MARYS HOSPITAL
- b.) Service ID (Figure 5-15, item b) is issued by the Service organization.
Example: 262785CT2 (no spaces)
- c.) Select the Time Zone for the site.

Note: Use the scrollbar at the bottom of the time-zone selection list to view the entire description of a time-zone, to ensure that you are selecting the correct time-zone for your location. If the time-zone of your location is not in the list, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See the LFC manual for more information regarding time-zone setting & selection.

- d.) Next MOD # *Customer selected.*
- e.) Next Patient Exam # configures the next Exam number the scan user interface will use. At initial system installation, type: 1
- f.) Next Diagnostic Exam # *Customer Selected.*

Note: Regenerate Database may not be present on your system. If it is not there, skip step g and go to step h.

- g.) Regenerate Database - Determines whether the Scan Database will be regenerated during reconfiguration. Select YES if this is an installation with no customer data present.

Note: If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database. This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See Load from Cold manual if you require more information regarding Scan Database Regeneration.

- h.) Mobile System Select to tell the software if this CT is in a mobile environment or not.

- 6.) Click the PREFERENCES button to display the Preference Settings Screen as shown in Figure 5-16.

Ask the customer which date and time formats to select.

Preferences Settings

Doctors Title: radiologist **a**

Units for patient weight: Pounds **b** Kilograms

Language: English **c** French Japanese German Italian Chinese Portuguese Spanish

AutoVoice Language: English-Male **h** French Japanese English-Female Italian Chinese German Spanish Korean Mexican Spanish

Date Format: MM/DD/YY **f** DD/MM/YY YY/MM/DD

Time Format: 12:00:00 AM/PM **g** 24:00:00 Military

Modified In Room Start: On **i** Off

HIPAA Present: On **j** Off

Dose Information Display: On **k** On without Total DLP Off

Select Preferred kVs (non-VCT Gantry ONLY) **d**
Note: All kVs are automatically enabled on VCT Systems
80 100 120 140

Target Noise Index Table **e**
Table 1 Table 2 Table 3

Figure 5-16 Preferences Setup Screen

- 7.) Configure Preferences Settings

- a.) Doctors Title: Enter the title for the Doctor (eg., radiologist.)
- b.) Units for Patient Weight: Tells the software whether pounds or kilograms are being used.

Note: Use Direction 5221102-1EN (found in the keyboard collector kit shipped with the system) to complete the language selection.

- c.) Language: configures the language to display on the Application screens.
- 1.) Review the language matrix (in direction 5221102-1EN) and identify the appropriate language for your country.
 - 2.) Ask the radiology manager, or equivalent, to agree to the user interface (UI) language and keyboard requirements listed for that country (per 5221102-1EN), OR choose another language for the UI and keyboard.
 - 3.) Record the information on GE form e4879.
- d.) Preferred Fast Cal KV: configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 80, 100, 120, 140. Use the default selection unless told differently.
- e.) Target Noise Index Table: Be sure Table 2 is selected.
- f.) Date Format: Configures the format in which the date will be displayed on the images.
- g.) Time Format: Configures the format in which the time will be displayed on the images.
- h.) AutoVoice Language: Configures the language heard in the scan room.
- i.) Modified In Room Start: Be sure "Off" is selected, unless the site is in Japan, in which case, this feature should be "On".

Comment:

These kVs should include all kVs that the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal kVs

- j.) HIPAA Present: Be sure Off is selected unless told differently.
- k.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose. Use the default selection unless told differently.
 - * Select ON (full CTDIw Display)
 - * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - * Select OFF (no CTDIw Display)
- 8.) Click the HARDWARE button to display the Hardware Settings Screen (Figure 5-17).

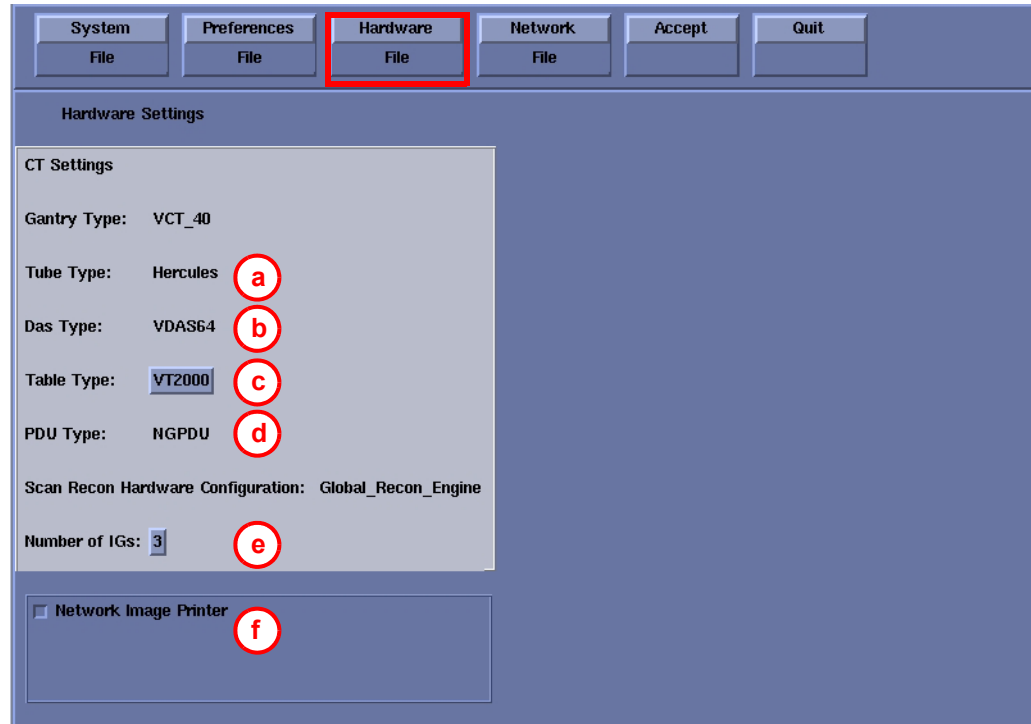


Figure 5-17 Hardware Settings Screen (Example only - Actual screen may vary)

- 9.) Configure Hardware Settings
 - a.) Tube Type Indicates the type of X-Ray Tube that is installed.
 - b.) DAS Type Indicates the type of DAS that is installed.
 - c.) Table Type Select the table type: VT2000 or VT1700. The type of table shipped can be determined by reviewing the order information.

Table Type on Configuration Screen	Table Type on Product Locator Card
VT2000	GT Long
VT1700	GT Short

Table 5-4 Table Type

- d.) PDU Type Indicates the type of PDU that is installed.
- e.) Number of IGs Choose the default, 3.
- f.) Network Image Printer Check this only if the site plans to use a network-connected Postscript printer (such as a Codonics printer) with this system.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e., a computer with a printer attached); a Network Image Printer (which is supported) is a printer attached directly to the network.

- 10.) Click the NETWORK button to display the Network Settings Screen as shown in [Figure 5-18](#).

Figure 5-18 Network Settings Screen

Comment: *This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.*
See [Chapter 7](#) for more information and complete details of setting the Hospital/System Network Configuration.

- 11.) Configure Network Settings:

- a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>_OC, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

- b.) Enter the hospital-provided Host Name.

The Host Name identifies the network hostname and AE Title of the CT system to the hospital's network. The Host Name:

- * **MUST NOT** exceed 16 characters.
- * **MUST** only contain the following characters: a through z, and 0 through 9.

The Host Name is typically stmary or ct01.

- c.) Enter the hospital provided IP Address for the system.

d.) The following procedure should be performed if your customer Ethernet hospital backbone begins with 172.16.0.xx, OR if the scanner will connect to a device with an IP address of 172.16.0.xxxx.

- * If the hospital backbone starts with 172.16.0.xx, and this is a first time install, change the VDARC subnet by checking the *Change DARC subnet* box and fill in the following value: 169.254.0. Continue and complete the software load on the HOST and VDARC.
- * If the hospital backbone starts with 172.16.0.xx, and the VDARC subnet was already changed, verify that the *Change DARC subnet* box is checked and a subnet is already filled. Continue and complete the software load on the Host and VDARC.
- * If the VDARC subnet was already changed and needs to be set to the 172.16.0 subnet, deselect the *Change DARC subnet* box. Continue and complete the load on HOST and VDARC.

e.) Enter the hospital provided Net Mask (if the CT system is on a subnet).

Note: If the hospital backbone IP address is 192.9.220.xx, then the Net Mask must be set to 255.255.255.252.

f.) Enter the Broadcast Address.

Comment:

The Broadcast Address should be the same as the IP Address except for the bits of the host ID portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other Sun OS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

g.) Enter the hospital-provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.

Note: Make sure the NUM LOCK button on the keyboard is not active. If it is, deselect it to enable the ACCEPT button in the next step. Failure to deactivate the NUM LOCK results in load issues.

h.) Verify with the customer's network administrator if NIS (sometimes called "Yellow Pages" database) is being used. If not, ensure that the ADVANCED OPTIONS and USE NIS boxes are not checked.

If NIS (commonly called "Yellow Pages" database) is being used, ensure that the ADVANCED OPTIONS and USE NIS boxes are checked and complete these steps:

- * Domain Name: Add the site's domain name here (Get the site's domain name from customer's network administrator.) (Example: CTSYSTEMS)
- * IP address of server: Add the IP address of the site's internal server here. (Get the IP address from customer's network administrator.)

12.) Review all screens to be sure the information entered is correct before proceeding to the next step.

13.) Click the ACCEPT button.

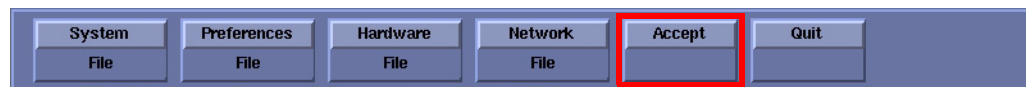


Figure 5-19 Accept Button

Comment:

The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a Shell window that closes when the loading process is complete.

- 14.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 5-20](#).)



Figure 5-20 Reboot Screen

- 15.) The system will automatically log in as `ctuser` after the reboot. Select OK on the Autostart Disabled popup message.
- 16.) Open a Shell window.
- 17.) Type: `st` ENTER

2.10 Set Time and Date

Verify that the date and time are correct.

- 1.) Open a Unix shell.
- 2.) Type `{ctuser@hostname} date` ENTER

If the date is correct, go to the next section, [Network and Service Integration and Checkout](#).

If the date and time are incorrect, continue with the following steps. You must set the time and date on the Host Computer with Application Software down.

- 1.) Open a Unix Shell and log in as root:
 - a.) Type: `su -` ENTER
 - b.) Type the root password and press ENTER
- 2.) Set the date and time for your time zone by updating the fields in the `setdate` routine.
 - a.) Type the following: `{root@hostname}# setdate`

Note: Type `q` to quit at any time; press ENTER to proceed.

To be accurate, this tool prompts you to enter the *Second*. Watch your clock or PC carefully to enter the proper value, and press ENTER at the right second to set the accurate time. Press ENTER to proceed.

- b.) Type the current Year (1980-2030).
 - c.) Type the current Month (1-12).
 - d.) Type the current Day (1-30).
 - e.) Type the current Hour (Military time, 0-23).
 - f.) Type the current Minute (0-59).
 - g.) Type the current Second (0-59).
- \Updating the time on the OC and DARC, Please Wait...
- Ping darc (172.16.0.2) 56(84) bytes of data.

2.11 Network and Service Integration and Checkout

- If additional network connections are needed for this installation, complete as required. Confirm network operation.
- If additional service integration is required to complete this installation, complete as required.
- Verify that the system information on the service home page is correct and that system service information is present on the service desktop.

2.12 Save System State

- 1.) Insert a new Save State DVD into the SCSI Tower DVD RAM drive.
- 2.) Click the SERVICE DESKTOP icon.
- 3.) If reloading software, click UTILITIES.

- 4.) Select SYSTEM STATE.



Figure 5-21 System State Save

- 5.) Click ALL to select all the cals, characterizations, etc.
6.) Click SAVE.
7.) If the following message appears, insert a DVD into the DVD drive and then click YES.



Figure 5-22 Save System State Prompt

- 8.) When completed, click DISMISS.
9.) Label and date the disk including the suite name.
10.) Close the Service Desktop window at the upper left corner of the screen.

2.13 Start Up Applications

Open the console Shell window. Type: `st` ENTER. The applications desktop appears on the monitor.

Section 3.0

Table Gantry Integration

3.1 Introduction

Use these procedures to functionally check every part of the table/gantry subsystem.

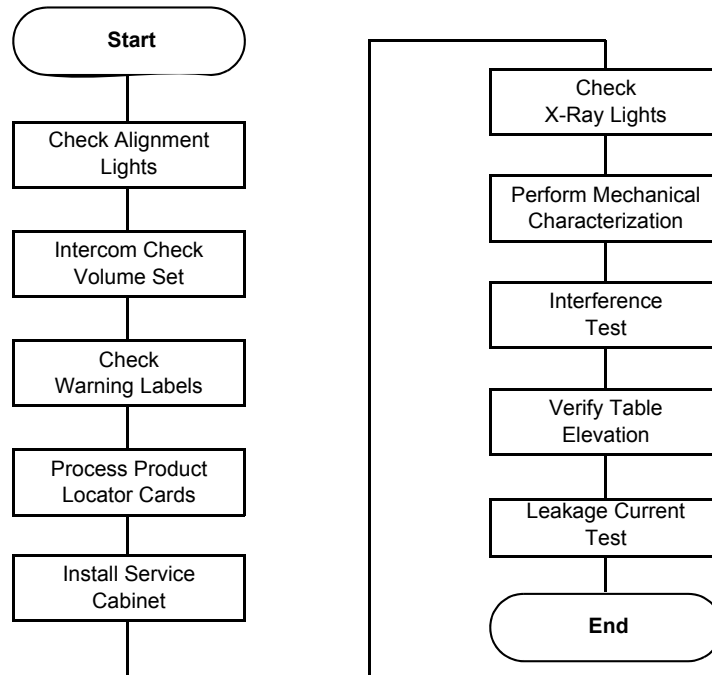


Figure 5-23 Table Gantry Integration Process Overview

Required Tool

- Multimeter

3.2 Check Alignment Lights

3.2.1 Room Light Adjustment

Adjust the scan room lights to normal customer operating levels.

3.2.2 Turning the Alignment Lights ON



CAUTION Verify all personnel have cleared the system. The Gantry rotates during this check.



- 1.) Turn ON the AXIAL DRIVE ENABLE and HVDC ENABLE switches (located on the service switch panel).
- 2.) Turn on the alignment light switch on the gantry service panel. The gantry will rotate and the alignment lights will turn ON.



CAUTION **LASER EYE INJURY!**

NEVER STARE DIRECTLY INTO THE LASER BEAMS WHEN YOU OPERATE THE ALIGNMENT LIGHTS. STARING INTO THE BEAMS CAN CAUSE PERMANENT EYE DAMAGE.

3.2.3 Internal Axial Lights

- 1.) Place a sheet of plain white paper over the output port of each light.
- 2.) Verify that the two laser lines coincide and appear as a single line.

Note: GE designed the internal axial lasers on the current CT system to shine *down* on the collimator. Do NOT adjust the internal alignment lights at this time. The tomographic plane tests use the QA phantom to check the internal axial lasers alignment to the collimator.

3.2.4 External Axial to Internal Axial Distance

Note: Ensure that cradle is level.

- 1.) Raise the table to its highest elevation.
- 2.) Extend the cradle until you see both the internal and external laser lights shining on the cradle.
- 3.) Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line. Verify this distance equals 240.0 mm +1.0 mm.
- 4.) Place the rule on the left edge of the cradle, and measure again.
- 5.) Leave the cradle in its current position, and lower the table to the minimum elevation.
- 6.) Measure the distance between the internal and external lights on both edges of the cradle, as above. Verify the distance remains equal to 240.0 mm +1.0 mm.

3.2.5 Coronal Lights

- 1.) Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light. Verify the two coronal lines coincide.
- 2.) Move the paper to the right side of the patient opening. Verify the two coronal lines coincide.
- 3.) Place the paper in the center of the gantry opening. Use a level to verify that the coronal lines are horizontal.

3.2.6 Turn Lights OFF

Press the alignment light button on the gantry control panel, again, to turn the lights OFF.

3.3 Autovoice/Intercom Checks

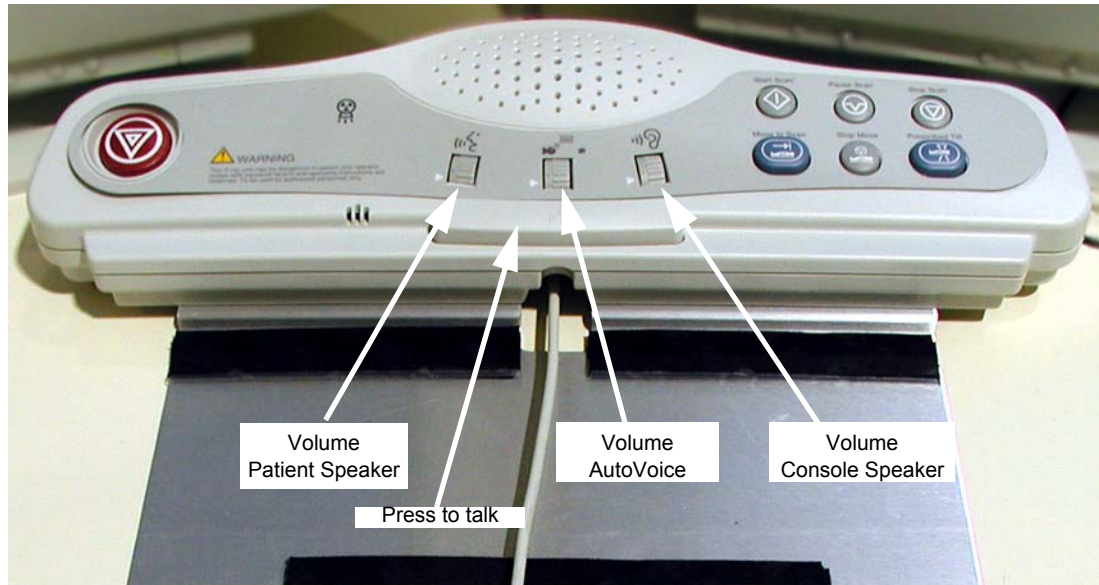


Figure 5-24 SCIM Volume Controls

3.3.1 Requirements

Two people are required to complete this procedure.

3.3.2 Patient Speaker

- 1.) To adjust the volume of the patient speaker in the table, adjust the left-most volume thumb wheel on the SCIM while speaking into the console microphone. (Press the bar on the SCIM to talk; release the bar to listen.)
- 2.) The patient should be able to clearly hear the operator.

3.3.3 Operator Console Speaker

To adjust the console speaker volume:

- 1.) Have an assistant speak into the gantry microphone.
- 2.) Adjust the SCIM console volume knob until you can clearly hear the patient.

3.3.4 Autovoice Volume

- 1.) On the Scan Desktop, select PROTOCOL MANAGEMENT.
- 2.) Select AUTO VOICE RECORD.
- 3.) Click the 3.4 button, to the right of "FF2. Inspiration".
- 4.) Click the PLAY button, to play the Inspiration AutoVoice message.
- 5.) Adjust the center volume thumb wheel while Autovoice is playing, to set the volume for the gantry speaker.
- 6.) Repeat steps 4 and 5 as necessary to achieve satisfactory volume.
- 7.) Select DONE, then select QUIT.

Note: If a satisfactory volume can not be achieved, refer to the system service manual and review the intercom module setup procedure.

3.4 CT System X-Ray ON Indicators, Cautions & Warning Labels

3.4.1 Check And Install System Warning Labels

All labels are installed in English and present on PDU, Console, Table, Gantry and Accessories. Replace the labels listed below ([Section 3.4.2](#) through [Section 3.4.7](#)) with the appropriate language labels for the country installed. Additionally, apply any other warning labels if present, on equipment where appropriate.

3.4.2 Console

3.4.2.1 On The SCIM

- Replace with the appropriate language overlay
- Replace with the appropriate language X-Ray warning label

3.4.2.2 On The Keyboard

- Replace with the appropriate language keyboard warning label
- Replace with appropriate language function key overlay label
- Replace with the appropriate language keyboard back warning label

3.4.3 Gantry

3.4.3.1 Scan Window Laser Labels

Replace with the appropriate language laser warning labels

3.4.3.2 Laser Window Labels

Replace with the appropriate language laser warning labels

3.4.3.3 Front Cover Laser Warning

Replace with the appropriate language laser warning labels

3.4.3.4 Front Cover Information Labels

Replace with the appropriate language labels

3.4.3.5 Gantry System GIB (Rating) label

Replace with the appropriate language system GIB label

3.4.4 Table - 2000mm

3.4.4.1 Front Side Cover - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.4.2 Back Cradle Pan - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.5 Table - 1700mm

3.4.5.1 Front Side Cover - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.5.2 Rear Side Cover - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.5.3 Back Cradle Pan - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.6 NGPDU

3.4.6.1 Front Cover - Emergency Off label

Replace with the appropriate language caution label

3.4.6.2 Front Cover - Gantry Enable Label

Replace with the appropriate language caution label

3.4.6.3 Front Cover - Power On Label

Replace with the appropriate language caution label

3.4.7 Accessories

3.4.7.1 Table Foot Extender Label

Replace with the appropriate language warning label

3.4.7.2 Coronal Head Holder Label

Replace with the appropriate language warning label

3.4.7.3 Sagittal Head Holder Label

Replace with the appropriate language warning label

3.4.7.4 Accessory Tray Label

Replace with the appropriate language warning label

3.4.7.5 IV Pole Label

Replace with the appropriate language caution label

3.4.8 Documentation - Verification

When finished update GE4879 (US Only) and the installation completion form that all appropriate language labels were installed and present.

3.5 Check Warning Labels

Note: The labels on the system and the system manuals must comply with the country law, as listed in Direction 5221102-1EN (found in the keyboard collector kit shipped with the system) regardless of the user interface (UI) language that is chosen. Compliance to the law must be completed prior to releasing the system to the customer.

Note: Do not cover English labels already on the system.

3.5.1 On SCIM

- 1.) Make sure the X-Ray warning label appears in the correct location on the SCIM.
- 2.) Record this information on Form 4879 located in [Chapter 6](#) of this book.

3.5.2 On Gantry

- 1.) Check that all laser warning labels are present on the gantry near the laser opening.
- 2.) There should also be warning labels on the lower right side of the gantry front cover.
- 3.) Record this information on Form 4879 located in [Chapter 6](#) of this book.

3.5.3 On Laser

- 1.) Make sure all laser warning labels appear in the correct location on the outside of the gantry.
- 2.) Obtain and install replacements for any missing labels.



Figure 5-25 Laser Warnings and Precautions

3.6 Process Product Locator Cards

- 1.) Collect the product locator cards shipped with the system. There should be approximately 28 product locator cards with the average system.
- 2.) Update the online product locator web site with the required hospital information.
- 3.) Confirm that the serial numbers on the cards shipped with the system match those found on the web site for that GON number.
- 4.) Update as required. Place the cards in a plastic bag, then place them in the service cabinet.

3.7 Install Service Cabinet

The service cabinet is shipped disassembled. The cabinet assembly process takes about 1.5 hours to complete.

- Note:**
- 1.) Assemble the cabinet following the instructions located in the cabinet's shipping box.
 - 2.) When completed, place the cabinet in the location shown on the site print.
 - 3.) Place all of the service materials shipped with the system in the service cabinet.

3.8 Check X-Ray Lights

Perform several scans following the steps below. Verify that the X-ray ON lights are ON during the scans. When done, check the boxes in [Table 5-5](#).

- 1.) Make sure the axial drive enable and HVDC enable switches are ON.
- 2.) If you are not on the Service Desktop, click on the Service Desktop icon.
- 3.) Select DIAGNOSTICS.
- 4.) Select DIAGNOSTIC DATA COLLECTION.
- 5.) Set the scan time to 2.00.
- 6.) Set the kV to 80.
- 7.) Set the mA to 40.
- 8.) Press ACCEPT RX.
- 9.) Press **START SCAN** button when flashing.
- 10.) Record the above information on Form 4879 located in [Chapter 6](#) of this book.

Light On?	Warning Light Locations
☐	SCIM
☐	Gantry Front
☐	Gantry Back (Use a mirror to view)
☐	Room Light (outside of the room)

Table 5-5 X-ray Light Chart

3.9 Mechanical Characterization

The relationship of table height to ISO center and internal-to-external landmarks must be characterized for proper interference matrix functionality.

Note: Do NOT perform tilt characterization.

3.9.1 Alignment Light Characterization

- 1.) Start the Mechanical Characterization tool from the Calibration tab on the Common Service Desktop.
- 2.) Select the CHARACTERIZE ALIGNMENT LIGHTS button from the interface.
- 3.) Follow the on-screen instructions.

3.9.2 Table Height Characterization

- 1.) Select the CHARACTERIZE TABLE HEIGHT button from the interface.
- 2.) Follow the on-screen instructions.

Note: If the table height is less than 21mm or greater than 25mm, relative to ISO, you must adjust the table height using the table leveling pad and adjusters. Raise or lower all four adjusters equally to achieve desired results.

3.10 Interference Test

PREREQUISITES

- Be sure that the System State was restored from DVD per Section [2.3 on page 191](#).
- Reset the hardware to download the new characterization values before performing the table/gantry interference tests in this section.

CONFIGURATION

- 1.) Table flashed with latest software.
- 2.) Table must have elevation and cradle and IMS characterized.
- 3.) Table must be mechanically aligned to gantry.
- 4.) Table must have had the table-gantry characterization completed.
- 5.) Verify the table extender is installed.

☐ Check box when complete.

TEST OUTLINE

The following tests verify the proper tilt and table interference matrix on the gantry.

- [Verify Table Elevation – Section 3.11](#)
- [Position Tilt, Move Table to Interference Limit – Section 3.11.1](#)
- [Position Table, Move Tilt to Interference Limit – Section 3.11.2](#)
- [Tilt Limits When Table Below Scan Plane Lower Limit – Section 3.11.3](#)

REQUIREMENTS

The following requirements are tested in this series of tests:

- 1.) No motion shall cause the table to hit the gantry (or gantry to hit the table)
- 2.) Requirement #1 shall include the use of the table extender.
- 3.) No tilt motion shall cause the gantry tilting frame to touch the stationary base covers for any tilt angle.

LIMITATIONS

These requirements will only be met when the table is NOT in service mode.

INTERPRETING TEST RESULTS

If test results indicate that elevation and/or tilt display readings fail to meet specifications, DO NOT adjust the limit switches. Instead, re-characterize and/or adjust tilt speed. Perform elevation and cradle and IMS first, then repeat the test. If it still fails, perform tilt.

3.11 Verify Table Elevation

#	TEST	EXPECTED RESULTS
3.11-1	Move the cradle (and IMS) to home position. Push the table down gantry push-button to lower the table to the minimum height.	Elevation Display should read 585.0 ± 3 mm
3.11-2	Raise the table to the maximum height using the gantry controls. If the mechanical alignment of the table/gantry is not correct, as is often the case during manufacturing staging, this value may be out of range. Most of the following tests will still be valid: Those that might have some variation are indicated in bold type .	Elevation Display should read 25.0 ± 3 mm. If mechanical alignment of the table/gantry is not correct, this value can be as low as 0 mm and as high as 40 mm.

Table 5-6 Table Elevation Tests

3.11.1 Position Tilt, Move Table to Interference Limit

The following tests verify the table interference limits at different tilt locations.

- Note:
- “S” means top of gantry tilts away from the table base.
 - “I” means top of gantry tilts toward the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table. Also, for all tilt angles used in this test, make sure that the Gantry Tilting frame covers do not touch the stationary base covers.

#	TEST	EXPECTED RESULTS
3.11.1-1	Move cradle (and IMS) to home position and set internal landmark.	Cradle position on display should read 0.0.
3.11.1-2	Raise table height to maximum height. Set the internal landmark, Move the IMS into gantry 400mm, as displayed on gantry. Set the internal landmark a second time. Move cradle into gantry 1m.	Table elevation on display should read 25.0 ± 3mm. (This value will be the same as in 3.11-2 in Table 5-6 above.) Cradle position on display should read 1000.0 mm.
3.11.1-3	Tilt the gantry to I30.0, then lower table until motion stops.	Table elevation on display should read 65 ± 3mm. Tilt display should read I30.
3.11.1-4	Tilt the gantry to I23.0, then lower table until motion stops.	Table elevation on display should read 124 ± 3mm. Tilt display should read I23.
3.11.1-5	Tilt the gantry to I20.0, then lower table until motion stops.	Table elevation on display should read 140 ± 3mm. Tilt display should read I20.
3.11.1-6	Raise the table elevation to maximum height.	Table elevation on display should read 25.0 ± 3mm. (This value will be the same as in 3.11-2 in Table 5-6 above.)
3.11.1-7	Tilt the gantry to S30.0, then lower table until motion stops.	Table elevation on display should read 147 ± 3mm. Tilt display should read S30.
3.11.1-8	Tilt the gantry to S23.0, then lower table until motion stops.	Table elevation on display should read 175 ± 3mm. Tilt display should read S23.
3.11.1-9	Tilt the gantry to S20.0, then lower table until motion stops.	Table elevation on display should read 185 ± 3mm. Tilt display should read S20.
3.11.1-10	Raise the table to 145 mm.	Table elevation on display should read 145 mm.

Table 5-7 Position Tilt, Move Table to Interference Limit Tests

#	TEST	EXPECTED RESULTS
3.11.1-11	Tilt gantry to S30 and verify the table height can be adjusted from 145 to 25 mm. (This value will be the same as in 3.11-2 in Table 5-6 above.)	Tilt display should read S30. Table lower limit should be 145 ± 3 mm. Upper table limit should be 25 ± 3 mm. (This value will be the same as in 3.11-2 in Table 5-6 above.)
3.11.1-12	Set the table height to 63 mm.	Table elevation on display should read 63 mm.
3.11.1-13	Tilt gantry to I30 and verify the table height can be adjusted from 63 to 25 mm. (This value will be the same as in 3.11-2 in Table 5-6 above.)	Tilt display should read I30. Table lower limit should be 63 ± 3 mm. Upper table limit should be 25 ± 3 mm. (This value will be the same as in 3.11-2 in Table 5-6 above.)

Table 5-7 Position Tilt, Move Table to Interference Limit Tests (Continued)

3.11.2 Position Table, Move Tilt to Interference Limit

The following tests verify the tilt interference limits at different table heights.

- Note:
- “I” means top of gantry tilts toward the table base
 - “S” means top of gantry tilts away from the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table.

#	TEST	EXPECTED RESULTS
3.11.2-1	Move cradle (and IMS) to home position and set internal landmark. Set gantry tilt to zero.	Cradle position on display should read 0.0. Gantry tilt on display should read 0.0.
3.11.2-2	Raise table height to maximum height, set the internal landmark, move cradle into gantry 1m.	Table elevation on display should read 25.0 ± 3 mm. (This value will be the same as 3.11-2 in Table 5-6 above.) Cradle position on display should read 1000.0 mm.
3.11.2-3	Lower table until height is 115mm. Tilt the gantry top away from the table (“S”) until it stops.	Table elevation on display should read 115 mm. Tilt display should read $S30 \pm 0.5^\circ$.
3.11.2-4	Tilt the gantry top toward the table (“I”) until it stops.	Table elevation on display should read 115 mm. Tilt display should read $I24 \pm 0.5^\circ$.
3.11.2-5	Tilt the gantry to 0. Lower table until height is 200mm. Tilt the gantry top away from the table (“S”) until it stops.	Table elevation on display should read 200 mm. Tilt display should read $S14.5 \pm 0.5^\circ$.
3.11.2-6	Tilt the gantry top toward the table (“I”) until it stops.	Table elevation on display should read 200 mm. Tilt display should read $I8 \pm 0.5^\circ$.
3.11.2-7	Tilt the gantry to 0. Lower table until height is 210 mm. Tilt the gantry top away from the table (“S”) until it stops.	Table elevation on display should read 210 mm. Tilt display should read $S10.5 \pm 0.5^\circ$.
3.11.2-8	Tilt the gantry top toward the table (“I”) until it stops.	Table elevation on display should read 210 mm. Tilt display should read $I6 \pm 0.5^\circ$.

Table 5-8 Position Table, Move Tilt to Interference Limit Tests

3.11.3 Tilt Limits When Table Below Scan Plane Lower Limit

The following tests verify the table and tilt interference limits when the table height is below the scan plane.

- Note:
- “I” means top of gantry tilts toward the table base
 - “S” means top of gantry tilts away from the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table.

#	TEST	EXPECTED RESULTS
3.11.3-1	Set gantry tilt to zero. Move cradle (and IMS) to home position, lower the table all the way, and set the internal landmark.	Cradle position on display should read 0.0 Gantry tilt on display should read 0.0. Table height should read 585 ± 3 mm. (This value will be the offset from 560 mm by the same amount as the upper table limit varies from 25 mm in 3.11-2 in Table 5-6 above.)
3.11.3-2	Tilt the gantry forward and backwards and verify the following tilt limits: S0.5 and I30.	Gantry tilt on display should read $S0.5 \pm 0.5^\circ$. Gantry tilt on display should read $I30.0 \pm 0.5^\circ$.
3.11.3-3	Set Gantry tilt to 0. Then, using the gantry push-buttons, move the cradle in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should stop at $10.0 \text{ mm} \pm 6 \text{ mm}$ from the home position. (This value will not be valid if the table/gantry mechanical characterization is not correct as described in 3.11-2 in Table 5-6 above. Insure that the front of the table is at least 25 mm from the gantry.)
3.11.3-4	Set gantry tilt to zero. Move cradle to home position.	Cradle position on display should read 0.0 Gantry tilt on display should read 0.0.
3.11.3-5	Raise the table to a height of 386 mm and verify the following tilt limits: S11.0 and I30.0.	Table height should read 386 mm. Gantry tilt on display should read $S11.0 \pm 0.5^\circ$. Gantry tilt on display should read $I30.0 \pm 0.5^\circ$.
3.11.3-6	Set Gantry tilt to 0. Set the internal landmark. Then, using the gantry push-buttons, move the cradle (and IMS) in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should stop at $141 \text{ mm} \pm 6 \text{ mm}$ from the home position.
3.11.3-7	Set gantry tilt to zero. Move cradle to home position.	Cradle position on display should read 0.0. Gantry tilt on display should read 0.0.
3.11.3-8	Raise the table to a height of 242 mm and verify the following tilt limits: S24.0 and I30.0.	Table height should read 242 mm. Gantry tilt on display should read $S24.0 \pm 0.5^\circ$. Gantry tilt on display should read $I30.0 \pm 0.5^\circ$.
3.11.3-9	Set Gantry tilt to 0. Raise the table to 210 mm. Then, using the gantry push-buttons, move the cradle in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should go all the way through the gantry bore to the full-extended position, approximately 2445 mm for GT 2000.
3.11.3-10	Set gantry tilt to zero. Move cradle (and IMS) to home position, lower the table all the way and set the internal landmark.	Cradle position on display should read 0.0 Gantry tilt on display should read 0.0. Table height should read 586 ± 3 mm. (This value will be the offset from 586 mm by the same amount as the upper table limit varies from 25 mm in 3.11-2 in Table 5-6 above.)
3.11.3-11	Tilt the gantry top toward the table to a tilt of 30 degrees	Display should read I30.
3.11.3-12	With the table down all the way, move the cradle in until it stops.	Cradle position should be 378.5 ± 3 mm.

Table 5-9 Tilt Limits When Table Below Scan Plane Lower Limit Tests

#	TEST	EXPECTED RESULTS
3.11.3-13	Move the table to the home position, raise the table to a height of 386 mm, set the internal landmark, and then move the cradle in until it stops.	Cradle position should be 403 ±5 mm.
3.11.3-14	Raise the table to a height of 63 mm, then move the cradle in.	Cradle should go all the way through the gantry bore to the full-extended position, approximately 2445 mm for GT 2000.

Table 5-9 Tilt Limits When Table Below Scan Plane Lower Limit Tests (Continued)

3.12 Leakage Current Test

Follow the instructions listed in [System-Level Safety Tests on page 265](#).

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Chapter 6

Image Quality




CAUTION

Shock Hazard.
Voltage Present.
No service on left side while energized.



NOTICE

Potential for Data Loss and/or Equipment Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 6](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Image Quality Process Overview Flowchart

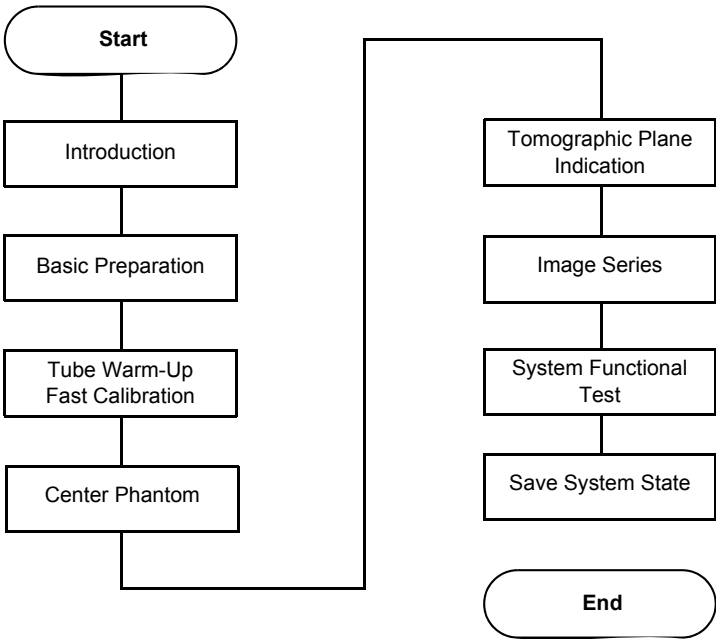


Figure 6-1 Image Quality Process Overview

Section 2.0

Reference Procedure - Scanning w/Service Protocols



NOTICE Do not perform the following procedure until instructed to do so in other sections of this chapter.

This procedure is used to locate the Manufacturing and Installation protocols on the SERVICE tab.

Note: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

PROCEDURE

- 1.) Select the NEW PATIENT icon on the left monitor.
- 2.) Enter a Patient ID (e.g., getest) for easy identification.
- 3.) Click on the box labeled SERVICE to access the necessary protocols.

Section 3.0

Tube Warm Up and Fast Cal

- 1.) Select DAILY PREP and TUBE WARMUP. Follow the steps on screen until all scans are completed.
- 2.) Select FAST CALIBRATION from the Daily Prep menu.

Four basic steps will be performed: Gantry Balance Check, Mylar Window Check (Warm up scans if tube is cold), Interconnectivity Map Scan List, and Fast Cal Air Scans.

Note: **Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)**

- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, click on QUIT.

Section 4.0

Center Phantom

REQUIRED TOOLS

- Standard FE Tool Kit
- 9" Level

PROCEDURE

- 1.) Locate the QA phantom and mount it and the phantom holder on the table.
- 2.) Use the adjustment knobs on the phantom holder to level the phantom front-to-back and side-to-side with a bubble level.
- 3.) Select SCANNER UTILITIES on the left monitor.
- 4.) Select CENTER PHANTOM.
- 5.) Follow the on-screen procedures.

Note: The phantom center spec is $\pm 0.5\text{mm}$.

- 6.) Select QUIT when the phantom is within specification.
- 7.) Level the phantom both front-to-back and side-to-side. (Use a 6" level.)

Section 5.0

Prepare the QA Phantom

Note: **The QA-3 phantom is shipped water-filled.**

- 1.) Locate the multi-language sticker packet in the QA phantom shipping box.
- 2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

Section 6.0

Tomographic Plane Indication

- 1.) Place the QA-1 phantom on the phantom holder.
- 2.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the line on the phantom lines up with the internal laser lights.
- 3.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
- 4.) Center the phantom in the scan plane with the CENTER PHANTOM found in the "Scanner Utilities" on the left monitor in the lower right corner.
- 5.) Click on CONFIRM.
 - a.) Adjust phantom as needed to achieve +/- 1.0 mm in both X and Y. Use CONFIRM to retry each time.
 - b.) Click DONE and QUIT when centered.
- 6.) Select NEW PATIENT.
- 7.) Select the service protocol, MANUFACTURING, TOMO PLANE INDICATION. The protocol should appear as shown in [Table 6-1](#).

Scan Type	Start Location	End Location	No. of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV	Recon Type
Helical Full 0.8 sec	I3.000	S3.200	32	0.625 10.62 0.531:1	0.200	S0.0	Small Body	120	100	2.1	25.0	Bone

Table 6-1 Tomographic Plane Protocol

- 8.) Select IMAGE WORKS and select Exam, Series, Image 1.
- 9.) Click on VIEWER.
- 10.) Select FORMAT and choose single image format view.

- 11.) Locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen. See Figure 6-2.



Figure 6-2 Exam, Series, Image 1

- 12.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 13.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0$ mm. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0$ mm requirement.
- 14.) Repeat the Tomographic Plane Indication test with the external alignment lights.
- Use the external alignment light, and press the external landmark.
 - Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
 - The scan plane indication must fall within the $S0.0 \pm 1.0$ mm specification.
- 15.) Initial below.

Section 7.0

Image Series

The Site Engineer will need to complete a full system detail calibration series in place of checking image quality.

If the customer requires IQ verification, follow the procedure in the customer's procedure manual, under the IQ section.

KVs/mAs and system alignment were completed during manufacturing system test, and calibration and file installation rechecks are not required.



NOTICE

It is recommended that you reboot your system before starting these calibrations.

From the operator's desktop, select SCANNER UTILITY. Complete the following three calibrations, in the listed order:

- 1.) Collimator Calibration
 - 2.) Detailed Calibration
 - 3.) Fast Calibration
- Exit. No other calibration steps are required.

Section 8.0

System Functional Test

Use the system tests in the following sections to exercise all aspects of the system and to ensure system integrity before releasing to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
 - Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.
- 1.) Place the QA-3 phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the line on the phantom with the internal laser lights.



NOTICE

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation-damaged detector, the cals may cause artifacts in subsequent image series scans.

- 2.) Select the service protocol SYSTEM SCAN/CUST QUAL RELI.
- 3.) Complete all scans.
- 4.) Review the message log for unacceptable errors and review the images for artifacts.

Section 9.0

Save System State

Use the following commands to create the System State DVD.

- 1.) Insert a DVD into the DVD drive.
- 2.) If you are not on the Service Desktop, click the SERVICE DESKTOP icon,.
- 3.) Click on the PM icon.
- 4.) Select SYSTEM STATE to open the System State Save/Restore menu.
- 5.) Select ALL
- 6.) Select SAVE
- 7.) When the save operation completes, select FILE and QUIT from the pull-down menu.
- 8.) Remove the DVD from the drive.

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Chapter 7

Customer Options Installation & Verification

Note: Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

CT Options Verification List

Complete the entire CT Options Installation Checklist and return to GE. The electronic file for the checklist is found on the LightSpeed 7.X Service Information CD-ROM.

NOTICE The following list should be used as a guide for *completion of actions only*. The electronic form contains additional fields. It **MUST** be completed in its entirety and returned to GE.

- ☐ Injector functional tests completed.
- ☐ AW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ Modem functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
 - * SmartScore Pro
 - * Smart Breath (Hardware installation required)
 - * CT Perfusion
 - * Cardiac (Hardware installation required)
- ☐ Teleradiology Connections (Refer to [Section 10.0 on page 257](#))
- ☐ Remote Monitor
 - * Auxiliary Monitor (Hardware installation required)
 - * Smart Step (Hardware installation required)
- ☐ Bar Code Reader

Section 2.0

Install Options

Refer to:

- GE Prints and schematics for mechanical (physical) location of option
- FDO shipment for identification of items
- Installation Specialist for installation instructions if they differ from print

2.1 Camera (Filming Device)

Refer to documentation shipped with camera

- Chapter 3 of this manual
- DICOM 2210573 GE Document
- DICOM Print 2152913

2.2 Injector

Refer to documentation shipped with the injector.

Complete a Functional test of the injector: The position of the ceiling mount assembly must allow the ceiling stand movement and gantry to tilt, typically 20 degrees, without interference.

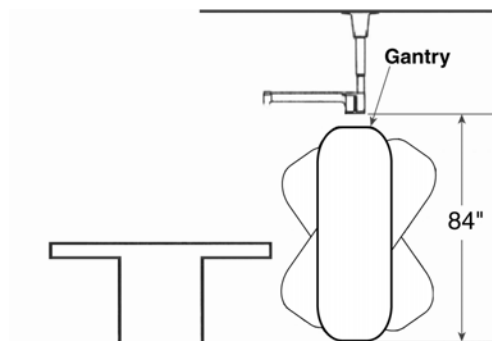


Figure 7-1 Ceiling Mount Position

2.3 Advantage Workstation (AW)

Refer to the directions provided:

- Pre-install 2111833
- Service 2111831

2.4 Power and Grounding Audit

A Power and Grounding Audit is required for this product.

2.5 Other Options

Refer to CT-2211224-100 *Lightspeed Options* for all other options.

Section 3.0

DICOM Network Introduction

LightSpeed systems support two basic Networking Protocols:

- Advantage NET (IC, Signa 4.X, CT-HLA, CT/I ...)
- DICOM (CT/I, CT Synergy, Advantage Workstations, ...)

DICOM NETWORK INTRODUCTION

DICOM networks basically operate on the **tasks** or services that various devices on the network use or provide. These services are labeled as Application Entity Titles (AE Titles). The CT scanner system is a **user** of six DICOM Network Services and is a **provider** of two DICOM Services:

AS A DICOM SERVICE USER:

- Send or **Push** images to another network device.
- Send or **Push** images to a DICOM Printer.
- Review image database on another device and retrieve or **Pull** selected images from that device (Query/Retrieve User).
- Send or **Push** images to a an image storage device and obtain confirmation that the images have been archived (Storage Commitment).
- Obtain Patient Worklist Information from the Hospital HIS/RIS System.
- Store images on MOD media.

AS A DICOM SERVICE PROVIDER:

- Receive **Pushed** images from another network device
- Allow another network device to review the image database and to retrieve or **Pull** selected images (Query/Retrieve Provider)

For each DICOM Service that the CT system will be a **User** (except for storing images on MOD media), you must **declare** this device on the CT system using three menu selections. For some devices, you must declare not only the device, but each service (AE Title) that the device provides. For example, you may be required to declare a PACS System twice on the CT system: once as a destination to **push** images and, second, as destination that provides storage commitment capability after images have been **pushed**.

For each DICOM Service that the CT system will be a **Provider**, you must **declare** the CT system on the network device that will be using these services.

Information required to complete configuring a hospital DICOM network is provided by the hospital network administrator (hostnames, IP Addresses) and the DICOM Conformance Statement document (AE Titles, Port Numbers) provided with each DICOM compatible network device on the network.

Section 4.0

Before You Start

4.1 Network Physical Requirements

Before setting up the CT scanner system on the hospital network, verify the following physical items are complete:

- Scanner console, monitor, keyboard, and mouse are installed and connected.
- CT system power is ON
- Hospital Ethernet network RJ45 Class IV twisted pair cable is connected to the scanner console network receptacle.
- Hospital network connection is operational and is running 10baseT or 100baseT.

4.2 Network Identity Information

To declare the CT system on the network, ensure the following network identity information is available from the Hospital Network Administrator:

- Hostname (No more than 16 Characters)
- Internet Protocol (IP) Address
- Subnet Net Mask IP Address (if applicable)
- Broadcast Address (if applicable)

4.3 Scanner to DICOM Remote Hosts Network Information

To declare DICOM remote hosts (PACS systems, archival devices, review workstations) on the CT system, ensure the following information is available for each remote host:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address
- Network Protocol (DICOM for CT Systems)

From the Remote Host Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

4.4 Scanner to DICOM HIS/RIS Interface Network Information

To declare DICOM Hospital HIS/RIS Interface devices (Mitra and others) on the CT System, ensure the following information is available:

From the Hospital Network Administrator: *Internet Protocol (IP) Address*

From the HIS/RIS Interface Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM Service that the HIS/RIS interface provides)
- DICOM Listening Port Number

4.5 Scanner to DICOM Printers Network Information

To declare DICOM on the CT System, ensure the following information is available for each printer:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address

From the Printer DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

Section 5.0

Declaring the System on the Hospital Network

5.1 Enter Configuration Routine

- 1.) On the OC, open a Shell window.
- 2.) Enter root as a superuser:
Type: `su -` ENTER at the prompt.
Type the password and press ENTER at the password prompt.
- 3.) Change directory to scripts:
Type: `cd /user/g/scripts` ENTER at the root prompt.
- 4.) Launch the Install Utility:
Type: `reconfig` ENTER at the prompt.
The OC displays the Install Utility Window as shown in [Figure 7-2](#).

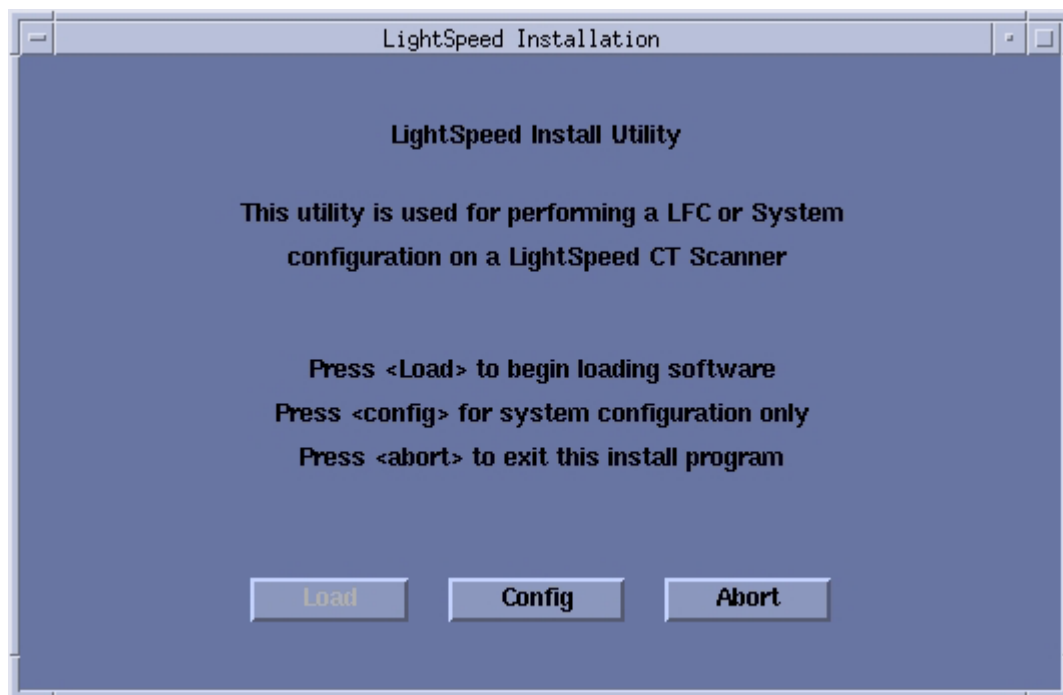


Figure 7-2 Install Utility Window

- 5.) Enter the Configuration Routine:

Using the mouse, click on the CONFIG button.

The OC displays the System Configuration - System Settings screen, as shown in [Figure 7-3](#).

System Settings

Hospital Name:

Service ID:

Next MOD #:

Time Zone:

Next Patient Exam #:

Next Diagnostic Exam #:

Regenerate Database?

☒ Yes ☐ No

Mobile System?

☐ Yes ☒ No

Location List:

- Africa
- America
- Arctic
- Asia
- Atlantic
- Australia
- Europe
- Indian
- Pacific
- Chicago
- Chihuahua
- Cordoba
- Costa Rica
- Cuiaba
- Curacao
- Danmarkshavn
- Dawson
- Dawson_Creek
- Denver
- Detroit
- Dominica
- Edmonton
- Eirunepe
- El Salvador

Figure 7-3 System Settings Screen

5.2 Configure Network Settings

This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 7-4](#).

Network Settings

Gateway Parameters

Suite Name:

Host Name:

IP Address:

Net Mask:

Broadcast Address:

Default Gateway:

Advanced Options

☒ Use NIS?

Domain Name:

IP address of Server:

☐ Change DARC Subnet?

Figure 7-4 Networks Settings Screen

- 2.) Enter the `Suite Name`.

The Suite Name is a means of identifying this particular CT system as a part of a group of CT Systems in a suite configuration. This Suite Name will appear on all image headers.

The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface will be `<Suite Name>_oc` and the SBC interface will be `<Suite Name>_sbc`.

- 3.) Enter the hospital provided `Host Name`.

The Host Name identifies the network hostname and AE Title of the CT system.

The Host Name:

- **MUST NOT** be `<Suite Name>_oc` or `<SUITE NAME>_OC`.
- **MUST NOT** exceed 16 Characters.
- **MUST** only contain the following characters: **A** through **Z**, **a** through **z**, **0** through **9**, **-** and **_**

- 4.) Enter the hospital provided `IP Address`.

- 5.) Enter the hospital provided `Net Mask` (if the CT system is on a subnet).

- 6.) Enter the `Broadcast Address`:

The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1s or 0s depending on the configuration of the network. The standard default is 1s but older SunOS machines used 0s.

Example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address.

If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

- 7.) Enter the hospital provided `Default Gateway IP Address` in the `Default Gateway` field (if applicable). If the site network does not use a default gateway, leave the field blank.
- 8.) Select NIS (Yellow Pages database) `Advanced Option` only if requested by the hospital network administrator as follows:
- a.) Select ADVANCED OPTIONS button on the Network Settings screen.
 - b.) Select Use NIS? button.
 - c.) Enter the hospital provided `Domain Name`.
- 9.) Record all the Network parameters in the *Software Installation Procedures* Document, or on the worksheet in Appendix E, [Section 4.0 on page 275](#).

5.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the System Configuration Screen.

The system loads the application software, OS patches, and kernel changes, and configures the system on both the OC and the SBC.

This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a Shell window, which closes when the loading process is complete. All the window output is logged to a file named:

`/var/adm/install.log.YYYYMMDDWWHHMMSS.`

(Where `YYYYMMDDWWHHMMSS` is the Date/Time that the loading process was started.)

- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES.
- 3.) The system will automatically login as `ctuser` after the reboot. Select OK on the Autostart Disabled popup message.
- 4.) To startup Applications, in the console Shell window, type `startup` ENTER.

Section 6.0

Declaring Remote Hosts on the CT System

6.1 Enter Remote Host Configuration Screen

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.

6.2 Declaring Advantage NET Remote Hosts on the Scanner

Use Advantage NET Protocol networks to communicate with older CT or MR Systems (MR Signa 4.x, CT-HLA, CT/I Systems, and Workstations that support the Advantage NET protocol). Advantage NET Protocol does not offer full compatibility with LightSpeed DICOM formats.

Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter Screen as shown in [Figure 7-5](#).

Figure 7-5 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).

- 4.) Select ADVANTAGE NET as the Network Protocol.
The system automatically removes the highlighting from the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.
- 5.) Select SAVE to store the parameter settings of the remote host.

6.3 Declaring DICOM Remote Hosts on the CT Scanner

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter screen as shown in [Figure 7-6](#).

Figure 7-6 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.
The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.

- 6.) Enter the **AE Title** from the DICOM Conformance Statement provided with the device.
Application Entity Titles (also known as ACR-Nema or Dicom Name) refer to the DICOM Network Services that a device provides to the CT System. For most devices, the AE Title is the same as the hostname (CT systems are equipped with this feature).
However, some devices such as PACS systems may have separate AE Titles and port numbers for each of the services that the PACS system provides. In these cases, you must enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services that the host provides (one host as an image **push-to** destination, another host as a **query/retrieve** provider, and another host as a **storage/commitment** provider).
Be sure to review the DICOM Conformance Statement for each device that will provide a remote host network service for the CT system (image **push-to** or store destination, Query/Retrieve, and Storage Commitment) to ensure that each service is correctly configured.
- 7.) Select the correct **Archive Node** choice for the device. The Archive Node selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider and indicate to the operator that a study/series/image was archived. Select:
 - **AUTO** to have the CT system automatically check to see if the designated remote host is a DICOM Storage/Commitment Provider.
 - **YES** if the device is the hospital designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen will indicate **Archive Status = Y** to the operator.
 - **NO** if the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct **Access to local host: settings**. These two selections allow you to selectively block the remote host from using the LightSpeed DICOM services as a provider (image **push-to** destination, and a **Query/Retrieve** provider).
 - **Send Images?** Set to **YES** if the customer wants the CT system to be able to have images **pushed** to the system from the applicable remote host. Set to **NO** if the customer wants to block an image **push** from the applicable remote host.
 - **Query/retrieve images?** Set to **YES** if the customer wants the remote host to be able to review the image database (query) and **pull** selected images from the database. Set to **NO** if the customer does not want the remote host to have this ability.
- 9.) Select the correct **Custom search?** setting. This selection allows the CT scanner to selectively search through the remote host's image database when the operator is using remote browser screen to **query** the remote host. The search parameters that the CT system allows the customer to use are: last name contains, patient ID, exam number, accession number, and exam date.
 - Select **ON** if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
 - Select **OFF** if the device does not support custom searches.
- 10.) Record all the remote host network parameters for each remote host in the *Software Installation Procedures* Document.
- 11.) Select **SAVE** to store the parameter settings of the remote host.

Section 7.0

Declaring the CT System on Remote Hosts

7.1 Declaring the Scanner on Advantage NET Protocol Devices/Systems

Refer to the appropriate service manual provided with the Advantage NET Protocol device or system to find instructions how to declare the CT System as an Advantage NET remote host.

7.2 Declaring the Scanner on DICOM Protocol Devices/Systems

Refer to the appropriate Service Manual provided with the DICOM protocol device or system to find instructions how to declare the CT System as a DICOM remote host.

The CT System provides two DICOM Services as a provider to remote hosts:

- A remote host can **push** images to the CT image database.
- A remote host can review the CT image database (query) and **pull** selected images (retrieve).

Use the following parameter information to configure the DICOM device/system to either **push** images to the CT scanner and/or perform a **Query/Retrieve** operation:

- **Hostname**: Provided by the Hospital Network Administrator. Exactly the same scanner assigned hostname entered in Network Configuration Screen.
- **Application Entity Title**: Exactly the same entry as the Hostname.
- **Network Address**: Provided by the Hospital Network Administrator. Exactly the same scanner assigned IP Address entered in Network Configuration Screen.
- **Network Protocol**: DICOM 3.0.
- **Port Number**: For all DICOM service that the CT System provides, use 4006.
- **Provider Type**: This field concerns the LightSpeed DICOM **Query/Retrieve** provider capability. All CT systems are study root systems, which allow queries at the exam, series, and image level.
- **Support Worklist**: This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All CT systems support a filtered search of the image database as part of the LightSpeed DICOM **Query/Retrieve** provider capability.

Section 8.0 DICOM HIS/RIS Setup

8.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the CT system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer. In addition, the CT system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

8.2 Loading ConnectPRO Software Option on the CT System

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.
- 2.) Click on the UTILITIES icon.
- 3.) Click on INSTALL.
- 4.) Select INSTALL OPTIONS and click START. The console displays the Software Options window as shown in [Figure 7-7](#).



Figure 7-7 Options Window when First Selected

- 5.) Insert the ConnectPRO Options MOD into the MOD drive.
- 6.) Click on OK to continue.
- 7.) On the Software Option window, select the CONNECTPRO option in the Available Options window.

- 8.) Click on the INSTALL button.

The console may display a message box during the software loading operation. When the system has completed loading the software:

- the Installed Options window displays the ConnectPRO option, and
- the console displays the ConnectPro Setup window as shown in [Figure 7-8](#).

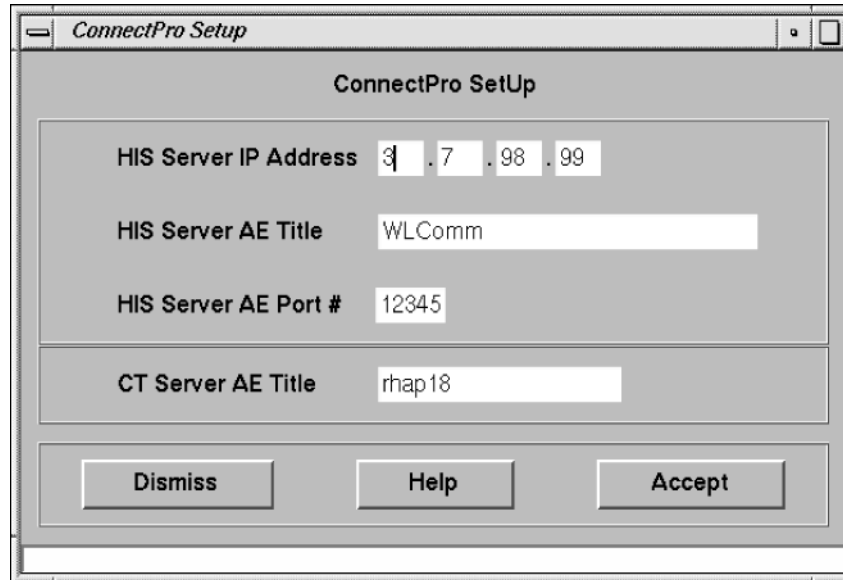


Figure 7-8 ConnectPro Setup Screen

- 9.) Enter the hospital provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the ConnectPRO HIS/RIS Network parameters in the *Software Installation Procedures* document.
- The system will automatically load the LightSpeed CT Server AE Title (Hostname of the system).
- 13.) Click on the ACCEPT button. The system will check valid setup entries, execute a perl script that makes changes to the configuration files for HIS/RIS, and return to the Software Option window shown in [Figure 7-7](#).
- 14.) Click on the QUIT button and the subsequent OK button in the message window.
- 15.) Remove the MOD from the MOD drive, and write protect the side containing the ConnectPRO option.

Section 9.0

DICOM Filming Devices Setup

9.1 Prerequisites

Before configuring DICOM filming devices (cameras, printers) on the CT System, ensure the following are complete:

- Filming Device Service Representative to assist in camera/printer setup for best image quality film presentation.
- Hospital DICOM network is operational.
- Filming device is connected to the DICOM network with the correct filmer DICOM interface.
- Filming device is DICOM protocol compatible.
- Filming device DICOM Conformance Statement document is available.

Note: **Filmer DICOM Application Entity Titles may be site specific. Make sure that you check with the Filming Device Service Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination filming device.**

9.2 Declaring DICOM Filming Devices on the CT System



NOTICE Potential For Data Loss

Empty all filming queues before modifying camera parameters.

This section contains procedures for recording important Camera setup information. Use the table(s) at the end of this section to record information from the setup screens.

- 1.) Click on the SERVICE DESKTOP button.
- 2.) On the Desktop Toolbar select UTILITIES -> INSTALL -> INSTALL CAMERA. The Install Camera window appears, along with a warning message pop-up box, to remind you that all filming queues must be empty before you begin to update or delete a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed (See Figure 7-10). First, you must click OK in the warning message box. See Figure 7-9.

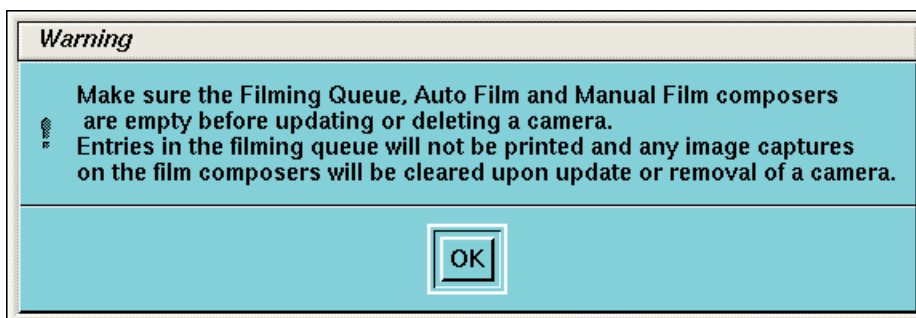


Figure 7-9 Warning Screen

- 4.) Now you are asked a series of questions.
- a.) To add a new camera, click the ADD button (See Figure 7-10).

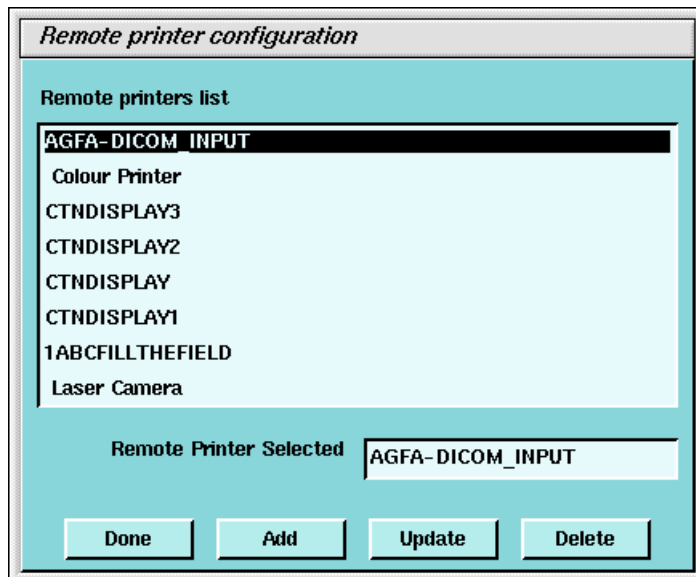


Figure 7-10 Printer Configuration GUI

- b.) A dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled (Figure 7-11). If a DASM is present and has not been detected, reboot the OC and run the camera configuration tool again.

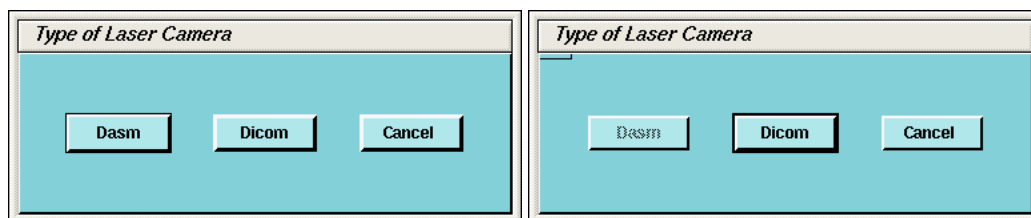


Figure 7-11 Dialog Box for camera Type (with and without DASM)

- 5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model from the list and click SELECT (See Figure 7-12). Now you must configure it.

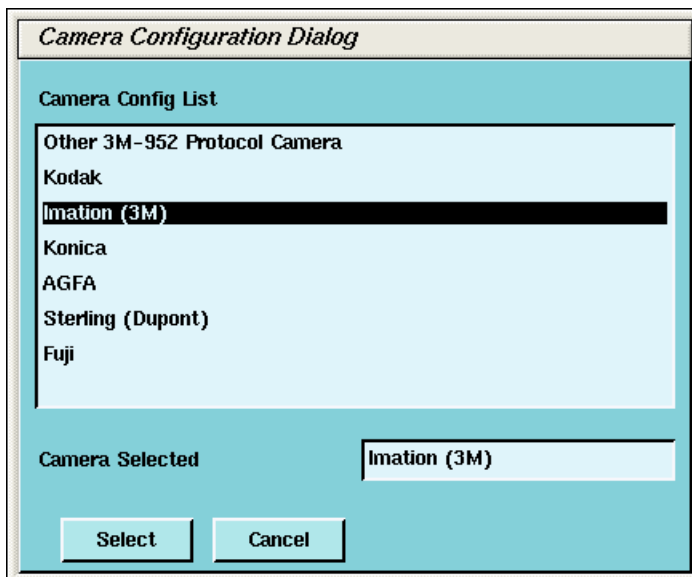


Figure 7-12 Camera Model Dialog (with DASM)

- a.) DASM Interface is automatically detected as either Analogue or Digital
- b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.
- c.) Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak: SMOOTH

Dupont/Sterling: SMOOTH

3M/Imation (Laser Camera): SHARP

3M/Imation (Dry View): SMOOTH

Agfa: SMOOTH

- d.) Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

When finished setting parameters, click on DONE and proceed to step 8.

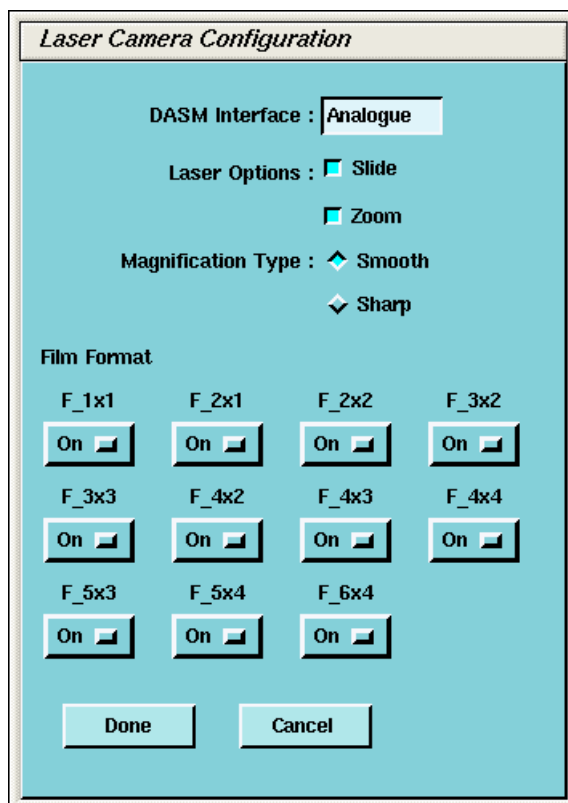


Figure 7-13 Laser Camera Configuration

- 6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.
- a.) A list of camera models appears (See Figure 7-14). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models except the Network parameters.

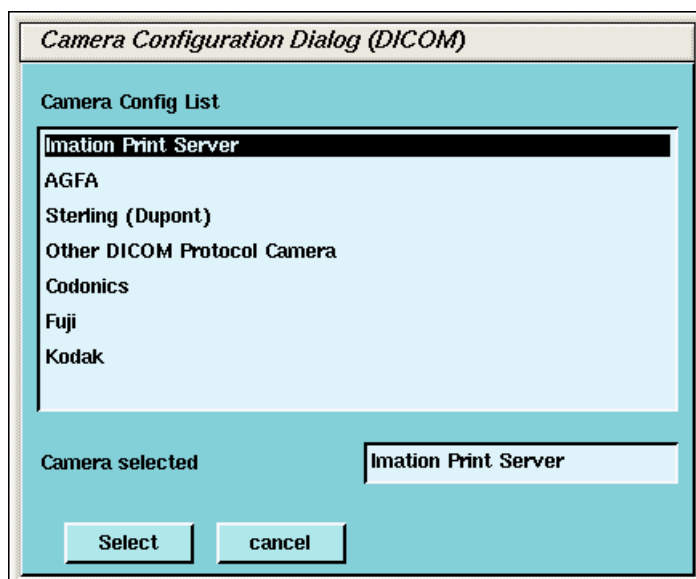


Figure 7-14 Camera Model Dialog (DICOM)

*Comment:
It's advised to
recheck the
preset
information with
the camera
manufacturer's
representative.*

Selection of a different camera model clears the Image Quality parameters, because these are camera manufacture dependent.

b.) Enter the Network Parameters (See Figure 7-15)

- > Device Name A unique name used to identify the camera.
- > Host Name DICOM print server host name, as defined by the hospital.
- > IP Address DICOM print server IP address, as defined by the hospital.
- > AE Title DICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure that you check with the Camera Manufacturer's Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- > TCP/IP Listen Port DICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturers DICOM Conformance Statement.*
- > Comments Optional comments used by the DICOM print server.

Figure 7-15 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and CLEAR FILM are supported.

- d.) Set Destination to the final location for film output: either MAGAZINE or PROCESSOR.
- e.) Orientation selects film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacture to ensure optimal image quality. Choices are describe below:
 - > None No interpolation. This option is not suported by all cameras.
 - > Replicate Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal preferred.*
 - > Bilinear A 1st order interpolation of pixels. Results in images usually described as blurred. *This algorithm is not normal preferred.*
 - > Cubic A 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing soefficients" is camera dependent.
- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-down menu box located below each selection. See Figure 7-15. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).
- h.) After the camera has been configured, click the ADVANCED button. This creates the camera device file for you automatically and pops up the Advanced Parameters screen. See Figure 7-16.

Comment: For most Camera Manufacturers, the preferred selection is CUBIC.

The screenshot shows a dialog box titled "Camera Attribute Dialog". It contains several sections of parameters:

- Image Quality Parameters**
 - Configuration:** A text field containing "PERCEPTION_LUT=KANAMC".
 - Maximum Density:** A text field containing "320".
 - Minimum Density:** A text field containing "32".
 - Smoothing Type:** A text field containing "140".
 - Empty Density:** A dropdown menu with "Black" selected.
 - Border Density:** A dropdown menu with "Black" selected.
- Film Size:** A dropdown menu with "<empty>" selected.
- Trim:** A dropdown menu with "Yes" selected.
- Priority:** A dropdown menu with "High" selected.

At the bottom of the dialog are two buttons: "Done" and "cancel".

Figure 7-16 Advanced Parameters (Camera Attribute Dialog)

7.) Advanced camera parameters control the image quality of films.

Note: For more information on the proper settings for these parameters, please see the Camera's DICOM Print Device Conformance Statement or the Camera Manufacturer Representative. You may need to refer to a copy of the Conformance Statement as you are working with the Camera Manufacturer's Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.

- a.) Configuration - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.

Note: Recommended Configuration Setting Values:

Agfa Drystar (MG3000)**PERCEPTION_LUT=KANAMORI (100)**

Imation Dryview (8700)**LUT=0,7**

Kodak Laser Printer 190**CS434\CN0\PD1.20**

- b.) Smoothing Type - Set Smoothing Type to ON, and input the selected value. This parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note: **Recommended Smoothing Type Starting Values and Ranges:**

Agfa DryStar (MG3000)Start Value:140Range:137 - 150

Imation Dryview (8700)Start Value:3Range:3 - 13

Kodak Laser Printer 190Start Value:EnhancedRange:Normal

- c.) Minimum and Maximum Density - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For Maximum Density, input the correct value into the text box. For Minimum Density, set it to ON and input the correct value in the text box.

Note: **Recommended Minimum and Maximum Density Starting Values:**

Agfa Drystar (MG3000)Min.:20 or 23Max:300

Imation Dryview (8700)Min.: (Blank) Max:300

Kodak Laser Printer 190Min.:20Max:300

- d.) Empty Image Density - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- e.) Border Density - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- f.) Film Size - Allows the system to specify a particular film size, if selected.
- g.) Trim - YES produces a white (clear) box surrounding each image.
- h.) Priority - This sets the print priority.
- i.) If you have completed entry of advanced parameters, click DONE.

CAMERA DATA TABLES: TO LOCATE INSTALL CAMERA INFORMATION: CLICK ON THE SERVICE DESKTOP BUTTON. ON THE DESKTOP TOOLBAR SELECT UTILITIES -> INSTALL -> INSTALL CAMERA. THE INSTALL CAMERA WINDOW APPEARS. SELECT EACH OF THE CAMERAS THAT ARE INSTALLED FROM THE LIST OF INSTALLED CAMERAS, AND CLICK ON UPDATE TO VIEW THE CAMERA'S SETTINGS. RECORD THE VALUES USED TO SET UP EACH CAMERA IN THE TABLES THAT FOLLOW. EXTRA TABLES ARE PROVIDED FOR MULTIPLE CAMERAS.

Note: You can determine this information by looking at the contents of the following files:

- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
- For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg` where, `name.cfg` is the camera device name from the printer configuration GUI.

Example: `more <filename from above> ENTER`

DASM Camera #1 Setup

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 7-1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-2 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-3 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-4 DICOM Camera #3

9.3 Troubleshooting Tips

The following is a summary of troubleshooting information for DICOM print that was gathered during software testing and validation of the DICOM print feature.

There is also a significant amount of additional troubleshooting procedures, and the theory of the DICOM print feature in the System Service Manual. Should you have problems installing a DICOM print camera, first read the information in System Service Manual.

ERROR BRINGING UP THE CAMERA INSTALLATION/FILMING APPLICATION

- **Symptom:** After creating/modifying the DICOM print camera the install camera interface does not come up and the filming application indicates it cannot bring up the film composer.
Cause: The configuration field contains invalid information such as a backslash (\) as the final character in the field or brackets ({}).
- **Solution:** The camera.dev file in `~ctuser/app-defaults/devices` must be manually edited to remove the offending characters in the set configuration line. Invalid characters include `\{ }`

CONFIGURATION INFORMATION FIELD

- **Symptom:** Cannot view the entire configuration field (> 25 characters)
- **Solution:** Hold down the middle mouse button and move the field contents

NEED TO SET DICOM PRINT ATTRIBUTES NOT SUPPORTED BY SOFTWARE

- **Symptom:** User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera.
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.

For Trim Off - set	TRIM NO
For Trim On - set	TRIM YES
- **Symptom:** DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for LightSpeed 7.X (14x17). [Otherwise the camera will queue the films or return an error causing the queue to pause (based upon the DICOM print camera specifications).]
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.

To force a 14x17 film size - set	filmSize 14INX17IN
----------------------------------	--------------------

NEED TO PREVENT DICOM PRINT ATTRIBUTES FROM BEING SENT TO DICOM PRINT CAMERA

- **Symptom:** Some DICOM print attributes are optional, and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.
- **Solution:** Using your favorite editor, add the following line(s) to the camera.dev file located in `~ctuser/app-defaults/devices` after the DICOM print device has been otherwise configured.
 - To prevent sending the Smoothing Parameter set `FB_Smooth FALSE`
 - To prevent sending the Border Density set `FB_Border FALSE`
 - To prevent sending the Empty Image Density set `FB_EID FALSE`
 - To prevent sending the Minimum Density set `FB_MinD FALSE`
 - To prevent sending the Trim Parameter set `FB_Trim FALSE`

ERROR TRYING TO CONNECT TO THE DICOM PRINT CAMERA

- **Symptom:** DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface" and the help message talks about running the `install.dasm` (Association Error)
- **Solution:** The system is unable to complete the association. Check the `AE Title` and the `Port number` of the DICOM print server and correct them through the Install Camera procedure.

FILM COMPOSER ERROR NOT USABLE

- **Symptom:** Film composer error says "unrecognized status - code 0"
- **Solution:** Review the log file, the attention and status windows. These areas have the correct filming status (for example, **film jam** and **supply empty**).

DEBUGGING CONNECTION ISSUES DIFFICULT

- **Symptom:** The timeouts for the DICOM print are very long, which means one needs to wait a long time before you know the application is not working.
- **Solution:** The timeouts for the DICOM print were setup to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print timeouts for the association and DIMSE classes (for example, **N-GET**, **N-DELETE**) can be modified within the DICOM print camera installation. They can be reduced down to 90 seconds.

DICOM PRINT ERROR ON N-GET TIMEOUT CONFUSING

- **Symptom:** When the N-GET timeout goes off, the error message in the `pr slog` file will be "Could not get printer status, invalid command sequence for N-GET".
- **Solution:** When the user sees the above error they may want to consider that the issue may be an inactivity timer on the N-GET DIMSE service.

DICOM PRINT CAMERA SLIDE SUPPORT

- **Symptom:** Current implementation of DICOM print does not allow selection of slide format.
- **Solution:** Feature not currently supported. Possibly in future releases.

CONFUSION ON FILM FORMAT NOTATION

- **Symptom:** GE Healthcare Laser Camera and DICOM Print film format notations are opposite.
- **Solution:**
 - GE Healthcare Laser Camera film format notation has always been row x col (for example, 12 on 1 = 4x3)
 - DICOM Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Section 10.0

Teleradiology (Framegrabber Type) Systems & Aux. Monitors

10.1 Introduction

Some systems may have teleradiology (TR) systems that framegrab the video. LightSpeed 7.X DOES NOT directly support this type of TR. The CT scanner's RGB color display video is a much larger format at a much higher pixel frequency. GE does not promise any direct compatibility with framegrabbing TR systems (DICOM 3.0 TR systems may work depending on the DICOM implementation but GE Healthcare does not and cannot validate all the various TRs.)

In the framegrabber case, a high quality (300Mhz bandwidth) video splitter/amplifier (as listed above) is needed to intercept and re-drive the display CRT RGB video. Composite grey-scale would then be available on Green #2 (1280x1024 pixels at 72Hz). Any framegrabber hardware attempting to capture this signal must be capable of a 140Mhz pixel rate. This also involves TR system configuration parameters. The TR capture software may also need upgrading to deal with 1280x1024 and/or **crop** the signal. The TR remote display software may need upgrading to view the larger format. The image transmission times to the remote TR may be up to 4 times as long. GE Healthcare will supply all technical information necessary to assist TR suppliers in making their systems work with LightSpeed 7.X, but GE Healthcare cannot be responsible for this third party TR equipment, software, or compatibility with LightSpeed 7.X.

The following common parts are available from GE Healthcare:

PART NUMBER	DESCRIPTION
2237018-2	Four-way Video Splitter (BNC Converter)
2154425	BNC to BNC 7' Splitter Cable
2142221	BNC to Host Computer 7' video Cable
2256482	DB15 to Host Computer 7' Video Cable
2256485	DB15 to BNC 7' Adapter Cable

Table 7-5 Common Parts Available From GE Healthcare

10.2 Auxiliary Monitors Setup

Figure 7-17 illustrates the **only approved GE CT configuration** for remote monitors, including the part numbers for the only approved components. The use of other configurations and/or components should be strictly avoided.

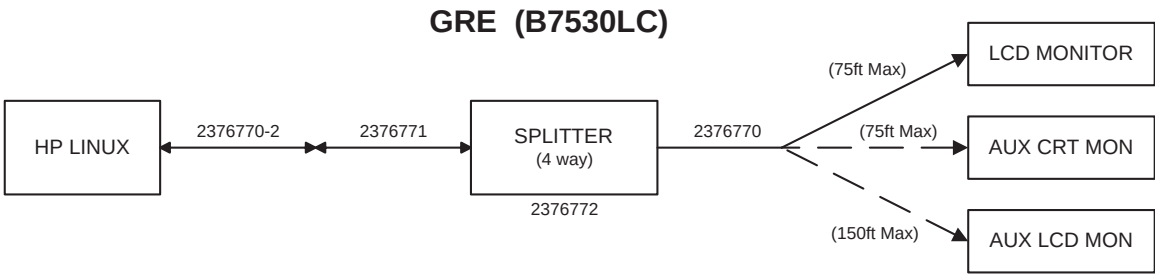


Figure 7-17 Video Splitter / Monitor Cabling

10.3 GE Healthcare CT Analog Filming Interface Specs (Video & Serial)

Note: The analog (VDB) video output specs and serial interface were specified from, and are the same as, the Genesis interface output specs. This does NOT apply to the digital (LCAM) filming interface.

OUTPUT	VALUE
amplitude	1 volt peak-to-peak
video	0.643V $\pm$ 10%
setup	0.071V $\pm$ 10%
sync	0.286V $\pm$ 10%
DAC resolution	8 bits
diff. linearity	$\pm$ 1 LSB max
glitch area	80 picovolt seconds max, for any step size
rise/fall times	> 10 nsec, $\pm$ 10-90%
FS settling time	7.5 nsec typical to 1 LSB
transfer function	guaranteed monotonic
noise level	> 5.0 millivolt peak-t~peak, combined sync/async noise
DC offset	$\pm$ 1 VDC referenced to ground

Table 7-6 Video Output (Measured Into 75Q at BNC Output)

OUTPUT	VALUE
logic family	F series TTL
output low level	0.8VDC max
output high level	2.0VDC max
output period	41.336 nsecs $\pm$ 10%
transition times	10 nsec max, $\pm$ 10-90%

Table 7-7 Pixel Clock Output

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
pixel frequency	24.192 Mhz	24.192 Mhz
pixel period	41.336 nsec	41.336 nsec
horizontal line frequency	33.6 Khz	33.6 khz
horizontal line width	720 pixels	720 pixels
horizontal active	544 pixels	544 pixels
horizontal blanking	176 pixels	176 pixels
horizontal front porch	26 pixels	26 pixels
horizontal sync	76 pixels	76 pixels
horizontal back porch	74 pixels	74 pixels

Table 7-8 Video Timing Characteristics

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
vertical frame frequency	60Hz	50Hz
vertical frame time	560 lines	672 lines
vertical active	524 lines	524 lines
vertical blanking	36 lines	148 lines
vertical sync	3 lines	3 lines
vertical back porch	30 lines	86 lines
vertical front porch	3 lines	59 lines
scanning format	non-interlaced	non-interlaced

Table 7-8 Video Timing Characteristics (Continued)

FORMAT	VALUE
visible field	544 pixels by 524 lines
image field	512 pixels by 512 lines
greyscale field	32 pixels by 16 level grey bar on left side of image
greyscale	software selectable on/off
greyscale off value	0 (black)
initial greyscale	255 (white) at upper left corner
border field	12 lines at bottom of visible field
border field value	any 8bit value, software programmable

Table 7-9 Video Display Format

ATTRIBUTE	VALUE
interface	R5422
25D conn pinout	pin 8 (RX+), pin 21 (RX-), pin 9 (TX+), pin 22 (TX-), pin 7 (GND)
baud rate	1200 baud
word length	8 bit, 1 start bit, 1 stop bit
parity	even
type	asynchronous

Table 7-10 Host Communications/Control Serial Port (analog interface only)

Note: The digital DASM/LCAM serial control has standard R5232 on pins 2, 3, & 7.

Section 11.0 Network Connections

BROAD-BAND

Broad-band is considered the standard network connection for LightSpeed 7.X. (A dial-up modem is optional.) Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the network through the Console.

- A patch cable (not to exceed 10 feet) should be provided by the customer, and it is used to connect the console to a wall box. (See Notes on [Figure 7-18](#))
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
- The run from the hospital switch to the CT wall outlet must not exceed 290 ft. (88m). Bandwidth performance is degraded when the length reaches 300 ft. (91m) or greater.
- For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice only line.



Figure 7-18 Console Rear Bulkhead

US PROCESS OVERVIEW

The United States network connectivity requirement for this product is broad-band. The US process relies on the Install Specialist to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

CUSTOMER BROADBAND RESPONSIBILITIES

Provide GE Healthcare Installation Specialist with an accurate site address, telephone number, contact name, and e-mail address for the:

- Customer Champion
 - Coordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments
 - Act as a focal point in assuring site broadband infrastructure meets GE Healthcare requirements for connection as determined by a mutual assessment with the GE Healthcare Connectivity team.
- IT Contact
 - Complete an equipment assessment with GE Healthcare Connectivity team to determine site readiness for broadband
 - Work with the Customer Champion to complete any identified infrastructure changes
 - Provide IP addresses for new CT equipment
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption
 - To utilize an Internet Service Provider that supports static routing



GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: _____ FE Name: _____
City, State: _____ FE Phone: _____
Date: _____ FE Email: _____

- | | Yes | No |
|---|---|--------------------------|
| 1. Does your site currently have a persistent (24x7) Internet connection? | ☐ | ☐ |
| 2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet? | ☐ | ☐ |
| 3. Does your site have a VPN device today? | ☐ | ☐ |
| 4. Is the VPN device one of the models below? <i>If Yes, please select the model from the options below.</i> | ☐ | ☐ |
| ☐ a) Cisco Pix Firewalls
☐ b) Cisco Routers
☐ c) Cisco 3000 Series (Altiga acquisition)
☐ d) Checkpoint Firewalls Software Version 4.1 and higher
☐ e) Nortel Contivity Software Version 3.2 or higher
☐ f) Redcreek | ☐ g) Symantec (Raptor) firewalls
☐ h) Firebox
☐ i) Linux S/WAN
☐ j) Sidewinder
☐ k) Netscreen
☐ l) None
☐ m) Other _____ | |

**If No, the GEMS Connectivity Support Team can help determine device compatibility.*

5. Does your VPN device support "triple DES" Encryption? ☐ Yes ☐ No
6. Has approval been given to install this VPN connection? ☐ Yes ☐ No
- Site Approver's Name _____
7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: _____
Installer Telephone Number: _____
Installer e-mail address: _____

Notes: _____

Field Engineer needs to provide compatible system information:

System ID	IP Address	Gateway Address
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

[Additional System and IP Address Spaces Available on Page 2](#)

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support 1-262-524-5261
Joseph.Gracz@med.ge.com

Once you have completed both pages of this form, please send it to:

a) Judy Heyer judy.hey@med.ge.com
b) Judy Heyer Fax# 414-918-4707

[Use the send button on page 2](#)



System ID List & Compatibility Matrix

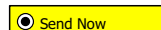
MR	Software Version	CT	Software Version
Include all Signa(LX, Ovation, OpenSpeed and CV/I) MR's System Ids and IP Addresses in assessment.		HiSpeed CT/e	Minimum 6.0
		HiSpeed CT/e dual	Minimum 2.06
		HiSpeedNX/I	Minimum 5.5
		LightSpeed QX/I	Minimum 1.3
		LightSpeed Plus	Minimum 2.1
Prior to completing an InSite checkout systems must be upgraded to Version 9.0 or 9.1 unless shown in the compatible list below.		LightSpeed Ultra	Minimum 3
Signa TwinSpeed	Release 9.0	HiSpeed X/I	
Signa	Release 9.1	CT/I	Minimum 6.2
Signa	Release 10.x	Lxi, Dxi, Fxi, Zxi	Minimum 6.01
Signa CV/I	CNV4		
Signa 3T			
Signa Profile	Minimum 7.66		
Signa Contour	Minimum 7.66		
Magnet Monitor	Minimum 2.3 software		
Network Products	Software Version	Nuclear/Pet	Software Version
AW 4.0	Version 4.0 or Above	Advance	Minimum 5.1
AW 4.0P		eNTEGRA	Minimum 2.03
X-Ray	Software Version		
<u>Cardiac</u>			
INNOVA	All		
<u>Mammography</u>			
Seno2000D	All		
<u>Digital Radiology</u>			
Revolution XQ/I	Version 10.12.5 or above		
Revolution XR/d	Version 18.0		
Multi Vendor			
MR	PSI	CT	PSI
Philips ACS 1.5T	MZP 400	Picker MX-TWIN	CZM 400
Philips NT 2000	MZP 401	Picker MX 8000	CZM 401
Philips NT 3000	MZP 402	Picker PQ	CZM 500
Philips NT ACS	MZP 403	Picker PQ 2000	CZM 501
Picker Outlook 0.23T	MZM 301	Picker PQ 5000	CZM 502
Picker Edge 1.5T	MZM 800	Picker PQ 6000	CZM 503
Picker Vista 1.0T	MZM 801	Picker PQS	CZM 504
Picker Eclipse 1.5T	MZM 900	Siemens AR STAR	CZS 100
Picker Polaris 1.0T	MZM 901	Siemens AR.C	CZS 101
Siemens Harmony 1.0T	MZS 100	Siemens AR.HP	CZS 102
Siemens Impact 1.0T	MZS 200	Siemens AR.SP	CZS 103
Siemens Impact Expert 1.0T	MZS 201	Siemens AR.T	CZS 104
Siemens SP 1.0T	MZS 400	Siemens PLUS-4 EXP	CZS 400
Siemens SP 1.5T	MZS 401	Siemens PLUS-4 POW	CZS 401
Siemens SP4000 1.5T	MZS 403	Siemens PLUS-4 VOL	CZS 402
Siemens Symphony 1.5T	MZS 500		
Siemens Vision 1.5T	MZS 600		

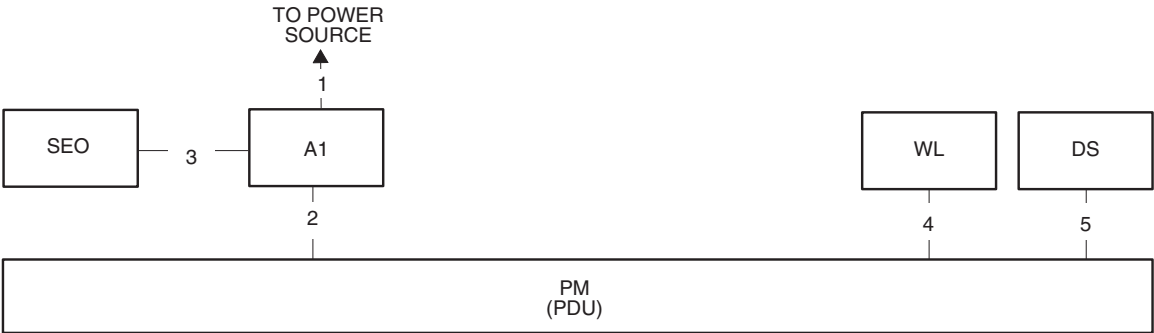
Field Engineer needs to provide compatible system information:

[illegible]

Once you have completed this form, please send it to:

Judy Heyer email **judy.heyer@med.ge.com**
Judy Heyer Fax # **414-918-4707**





- NOTES:**
- 1) Used for remote diagnostics - Option
 - 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
 - 3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

Only one phone connection is required for the system.

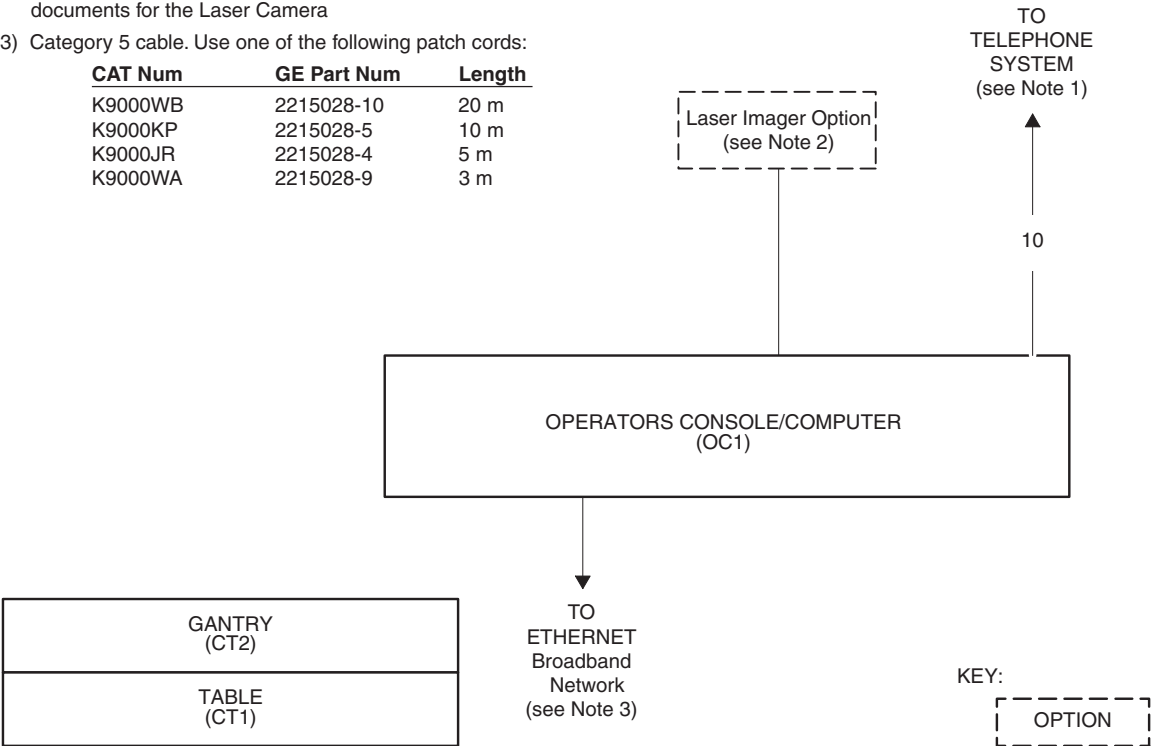


Figure 7-19 LightSpeed System Interconnection Runs

Chapter 8

System-Level Safety Tests

Section 1.0

Patient Touch Leakage Test

1.1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	10 mins	20 mins	10 mins

1.2 Overview

This procedure must be completed after all options have been installed. It covers three safety and leakage current checks:

- Patient Touch current test (completed after installation)
- System Ground resistance measurement (completed during installation)
- Ground current typical (completed after installation - optional)

1.3 Preliminary Requirements

1.3.1 Tools and Test Equipment

Item	Qty
Standard service tool kit	1
Dale 600 Meter (from Tool Pool) (P/N 46-328406G1)	1
Dale extended length leads (Part of P/N 46-328406G1)	1

1.3.2 Safety



WARNING



POTENTIAL FOR SHOCK

GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER “ON”. FOLLOW APPROPRIATE SAFETY PROCEDURES FOR WORKING WITH AN ENERGIZED SYSTEM.



NOTICE

Follow ALL required safety and PPE procedures customary for your organization, when working on this product.

1.3.3 Required Conditions

- Only trained service personnel should service the GE CT Scanner.
- Footswitch cover must be removed.

1.4 Procedure

1.4.1 Patient Touch Leakage Current Test Procedure

- 1.) Move the table to ISO elevation.
- 2.) Remove the footswitch covers and the gantry left side cover.
- 3.) Refer to the Dale 601 Operator's Manual for instructions on the use of the Dale 600 meter for measuring leakage current (or refer to [Figure 8-1](#), for a quick overview)

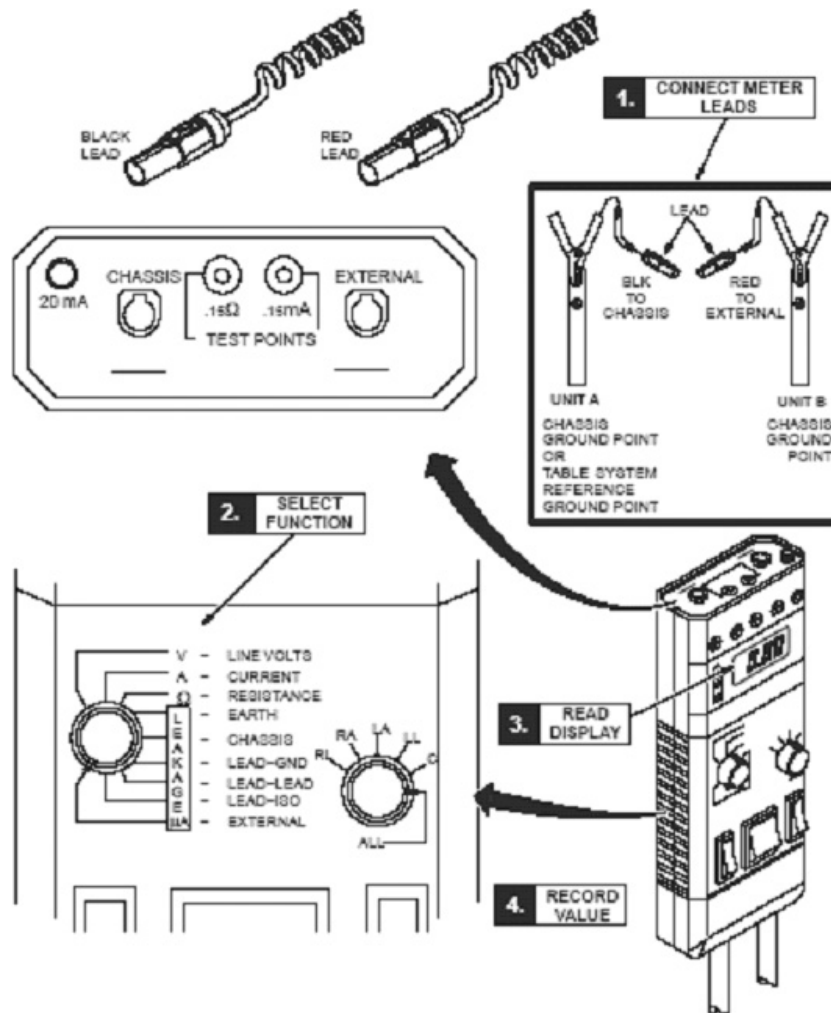


Figure 8-1 Using Dale 600 to Measure Leakage Current

- 4.) Plug the Dale 600 Meter into the outlet on the gantry left side. Confirm the outlet is correctly wired per the three LED indicators on the meter.
- 5.) Connect one end of the shorter black lead to the chassis plug and connect the other end to the table ground bus.
- 6.) Connect the longer red test lead (or the longer black lead) to the external plug on the top of the Dale 600 meter.

- 7.) Set the function switch on the Dale to external. Use the external lead to touch the meter's test terminal, to test that the meter is operational.
- 8.) The black lead will be connected to the table base ground bus, and the read lead will be connected to the devices (components) under test.
- Note: Your meter may have two black leads that are keyed for chassis and the ground connection. A valid calibration sticker must be present on the meter used. Record this information on the GE 4879 form.
- 9.) Beware: Leakage current cannot exceed the following and is tested with power ON:
- a.) **Critical care areas (invasive) - 10µA CT Systems**
 - b.) General care areas - 20µA
 - c.) Not intended for patient area - 50µA
- 10.) Testing must be completed between the system reference ground point (table base) to unit reference ground points (i.e. gantry and table, see [Figure 8-3](#)).
- 11.) Test all conductive surfaces and components within patient reach or within 6 feet (1.9m) of the table and 7.5 feet (2.3m) above.
- Measure at table maximum travels.
 - At some sites, wall outlet cover plates and sinks may become an issue.
- 12.) Test all Optional components such as in-room monitors, injector, overhead monitor suspensions, and table options.
- Record these results in the data sheet. To be completed at install and during PM inspections.

Gnd Bus to	Install <10µA
Any table ground points within the 6' range	
Any gantry ground points within the 6' range	
Injector assembly metal surface	
Boom-in-Room metal surface	
Monitors or metal surface	
Sink or metal surface	
Installed table accessories	
(other)	
(other)	
(other)	

- 13.) **Test conditions:**
- a.) Table max elevation Min - Max travel
 - b.) Patient face up and face down head first feet first
 - c.) Table min elevation 7.2' patient access to all conductive surfaces including:
 - * IV poles and tray assembly
 - * any smart step monitors and stands
 - * table bearing rails
 - * all other conductive surfaces



NOTICE

Be Aware of Static Discharge from: scan window, keypads, display, and touch pads or other plastic surfaces.

14.) Procedure Hints:

- a.) Look for items that have abnormal measurements, high or low.
These could indicate mis-wiring, loose connections, or poor connections due to corrosion, painted surfaces, etc.
- b.) High leakage could indicate a wiring error such as a neutral connected to the ground.
- c.) Fluctuating ground currents could indicate a short, poor connection, or facilities ground problem causing leakage currents from other areas of the facility to flow through our system grounds.
- d.) Refer to the illustrations below, for measurement visual descriptions.

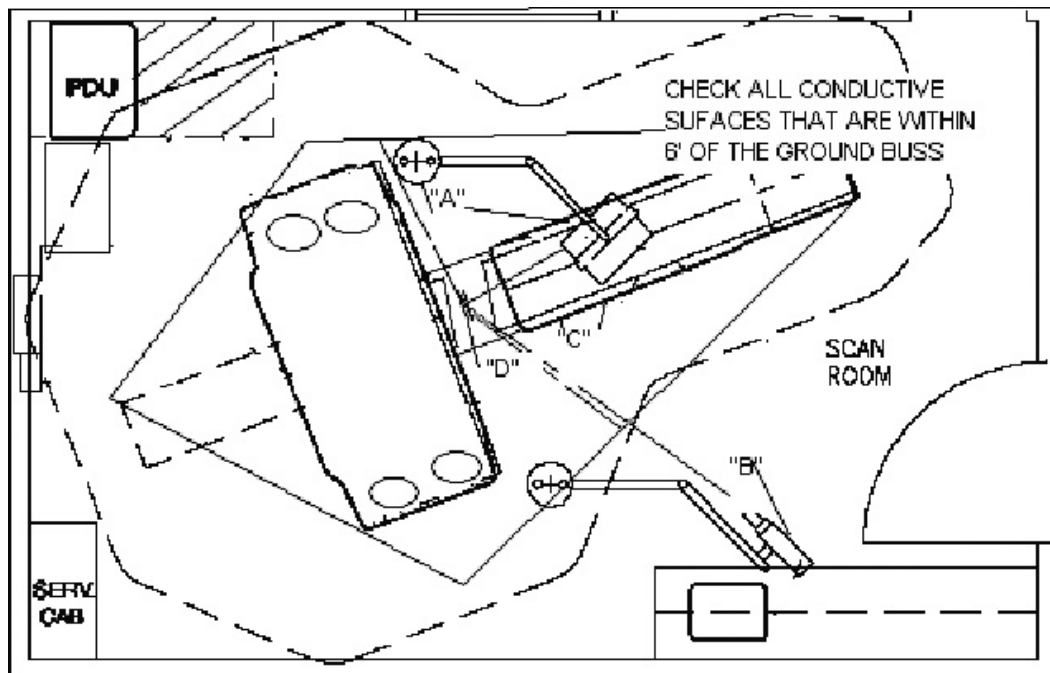


Figure 8-2 CT Leakage Current Map

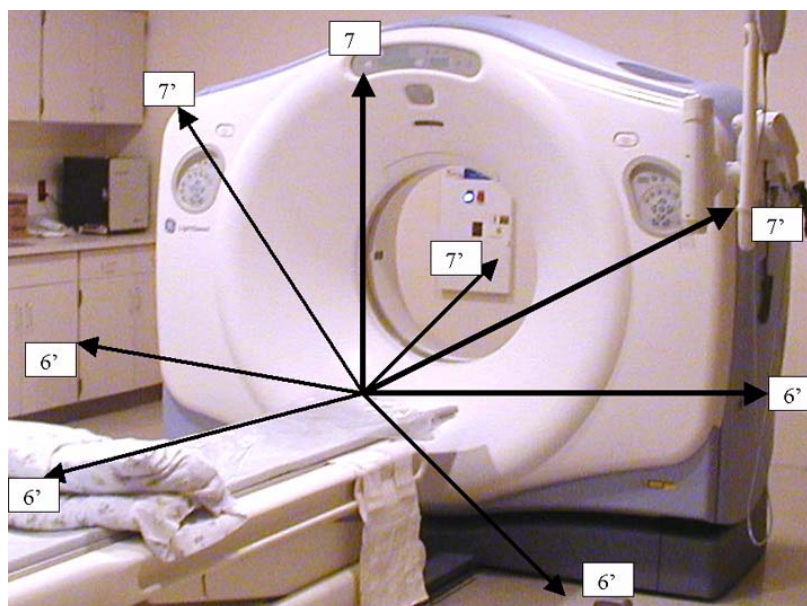


Figure 8-3 Measurement Points

1.5 Ground Resistance Checks Procedure (done during mechanical install)

Refer to the procedures found in Book 1, Chapter 3:

- Section 1.0 - System Continuity & Ground Checks (Mechanical Contractor)
- Section 2.0 - Site Ground Continuity Check

1.6 Finalization

No finalization required.

Section 2.0

CT System Chassis Leakage Test

2.1 Personnel Requirements

- 1 Engineer
- 1 Customer electrician

2.1.1 Time

2 Hour labor on site, includes:

- Removal of covers
- Leakage test
- Reinstallation of covers
- Recording of site data on the CT System Chassis Leakage Test Completion Form, if required by your state. Forward the completed form to your Project Manager of Installation.

2.2 Preliminary Requirements

2.2.1 Tools and Test Equipment

Item	Qty
Standard FE Service Tool Kit	1
GEH LOTO Kit	1
Dale 600 or 601 Leakage Current Meter (from Tool Pool) (P/N 46-328406G1)	1
*This procedure has been validated only with the Dale 600 / 601. GE Healthcare cannot guarantee the accuracy of this procedure if you use another meter.	

2.2.2 Safety

-  **DANGER** **POTENTIAL FOR ELECTRIC SHOCK**
GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER “ON”.
THIS TEST IS PERFORMED WITH POWER “ON” - SYSTEM IN STAND-BY!
-  **NOTICE** Follow ALL required safety, PPE, and arc-flash procedures customary for your organization, when working on this product.

2.2.3 Required Documentation

Refer to Equipment Service section – LOTO PPE for detailed procedure.

-  **DANGER** **POTENTIAL FOR ELECTRIC SHOCK**
SERVICING HARDWARE WITH POWER “ON”.

2.3 Procedure

Follow LOTO and other safety procedures found in this manual before starting this procedure.

- 1.) Remove the Gantry top and side covers. Guide rails on each side of the top cover will aid in removal. Refer to Appendix A: [Removal & Installation of Covers on page 123](#) for more details.
 - Note: Fans located on the top cover will make that cover heavier.
 - 2.) Power down the console and follow the GEH LOTO procedure.
 - 3.) Remove the Gantry front cover, following the procedure in [Appendix A on page 123](#).
 - 4.) Remove the Table foot switch top cover to gain access to the ground cables and ground bar.
 - Note: Do NOT disconnect ANY grounds at this time.
 - 5.) Confirm that all system grounds are securely attached to the system ground buss and NOT the Table base.
 - 6.) The electrician will remove all external electrical connections made during installation, including:
 - a.) Main system ground at PDU
 - b.) Power feeder flex connection at PDU
 - c.) Room door interlocks and room warning light connections
 - d.) Any and all other external ground connections to the system.
 - 7.) Confirm that all external gantry, table, console, and PDU connections have been removed.
 - Note: Some wires such as the room warning light may have external power and wire nuts, which should be installed to protect from arching.
 - 8.) Follow the LOTO procedure for re-energizing power and boot to application level. If not already done, remove the footswitch assembly top cover while the system boots.
 - 9.) Plug the Dale 600 / 601 Leakage Current Meter into one of the outlets on the gantry.
 - 10.) Connect the meter leads to the meter as follows:
 - a.) Connect one end of the shorter black lead to the chassis plug and the other to the table ground bus.
 - b.) Connect the longer red test lead to the external plug on top of the Dale 600 / 601 meter.
 - 11.) Set the function switch on the Dale 600 / 601 meter to EXTERNAL. Using the external lead, touch the meter's test terminal to confirm that the meter is operational.
 - Note: For more detailed information, refer to the Dale 600 / 601 Operator's Manual or see [Figure 8-1](#), for a quick overview.
- With the system at application-level and all components functional, test the system ground wires as follows:
- a.) Remove a system ground wire.
 - b.) Test that system ground wire.
 - c.) Replace that system ground wire.
 - d.) Repeat, testing all system ground wires ONE-AT-A-TIME. A list of the system ground wires appears in [Table 8-1 on page 272](#).
- Note: The measured leakage current must not exceed 5 MA in any ground wire

Components	Results	Date	Tester
PDU Leakage 1/0			
Gantry Leakage 1/0			
Console Leakage #2			
Table Leakage #10			
Option Grounds, if present			

Table 8-1 System Ground Measurements

- 12.) After completing all tests, follow the LOTO procedure to power down the system.
- 13.) The electrician will reinstall all electrical connections, conduits, cables, and wires removed in Step 6 and will secure all connections per NEC code. Check that all connections are securely tightened.
- 14.) Reinstall all removed system covers, except for the gantry right-side cover, located by the service switch panel.

2.3.1 System Power-Up and Test

Note: Complete this section of the installation manual on-site.

- 1.) Follow the LOTO procedure for re-energizing power.
- 2.) Turn-on the gantry service switches and power up the console.
- 3.) Check that no cables remain in the gantry rotating path.
- 4.) Return cover dollies to storage areas.
- 5.) Check that the table controls and footswitch function properly.
- 6.) Re-test the system by completing a system functional scan. If installed, be sure to test the room warning light and the door interlock at this time.
- 7.) Complete the CT System Chassis Leakage test, if required by your state, and forward the completed form to your Project Manager of Installation. (This form is located on the Service CD.)

2.4 Finalization

No finalization required.

Appendix E

System Configuration Data Sheets

Section 1.0

Requirements

- Record valuable system information in the data sheets that follow. Consult with your customer or network administrator to obtain the information. Understanding how the customer plans to use their CT scanner and their network and filming expectation reduces the time required to reconfigure the system.
- [Table E-1 Manual Film Composer Options, on page 273](#)
 - [Table E-2 System Network Configuration, on page 274](#)
 - [Table E-3 Networking Application \(Image transfer\) Configuration, on page 275](#)
 - [Table E-4 DASM Laser Camera Configuration, on page 275](#)
 - [Table E-5 DICOM Print Camera Configuration, on page 276](#)
 - [Table E-6 DICOM Print Camera Advanced Configuration, on page 276](#)

Section 2.0

Manual Film Composer Options

MANUAL FILM COMPOSER OPTIONS	
Slide Format (if available):	
Greyscale:	
Auto Printing:	
Auto Clear Page:	
Icon Labels:	
Expose Order:	
No. of Copies:	

Table E-1 Manual Film Composer Options

Section 3.0

System Network Configuration

SYSTEM NETWORK CONFIGURATION			
	FIELD NAME:	SETENV NAME:	FIELD VALUE:
System Settings:	Service ID	SERVICE_ID	
	Hospital Name	HOSPITAL_NAME	
	Exam Number *	* Ask customer or check log	
	DAS Type	DASTYPE	
	PDU Type	PDUTYPE	
Network Settings:	Gateway Host Name	GATEWAY_HOSTNAME	
	Gateway IP	GATEWAY_IP	
	Gateway Net Mask	GATEWAY_NETMASK	
	Gateway Broadcast Mask	GATEWAY_BROADCAST	
	Suite Name	SUITEID	
Option	Network Printer IP Address		
Option	HIS Server IP Address		
Option	HIS Server AE Title		
Option	HIS server AE Port		
Option	CT Server AE Title		
Option	Connect Pro IP Address		

Table E-2 System Network Configuration

HOST ETHERNET ADDRESS

_____ : _____ : _____ : _____ : _____

Section 4.0

Network Application (Image Transfer) Configuration

Record the network application (image transfer) configuration.

NETWORKING APPLICATION (IMAGE TRANSFER) CONFIGURATION				
AE TITLE OR HOST NAME	NETWORK ADDRESS	NETWORK PROTOCOL	PORT NUMBER	COMMENTS

Table E-3 Networking Application (Image transfer) Configuration

Section 5.0

Camera Application Configuration

Record the camera application configuration for the DASM or DICOM print camera.

DASM LASER CAMERA CONFIGURATION	
Camera Type:	
DASM Type:	
Film Smooth/Sharp Setting:	
Options:	
Valid Film Formats:	
Default Film Formats:	

Table E-4 DASM Laser Camera Configuration

DICOM PRINT CAMERA CONFIGURATION

Camera Type:	
Host Name:	
IP Address:	
AE Title:	
TCP/IP Listen Port:	
Comments (Optional):	
Valid Film Formats:	
Default Film Formats:	
Destination:	
Orientation:	
Medium Type:	
Magnification Type:	

Table E-5 DICOM Print Camera Configuration

DICOM PRINT CAMERA ADVANCED CONFIGURATION

Smoothing Type:	
Configuration:	
Minimum Density:	
Maximum Density:	
Empty Density:	
Border Density:	
Association Timeout:	
Session Timeout:	
N-Set Timeout:	
N-Action Timeout:	
N-Create Timeout:	
N-Delete Timeout:	
N-Get Timeout:	

Table E-6 DICOM Print Camera Advanced Configuration

Appendix F

Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5032	Alternating Current
	335-1	Three-phase Alternating Current
	335-1	Three-phase Alternating Current with neutral conductor
		Direct Current
	417-5019	Protective Earth (Ground)
	348	Attention, consult ACCOMPANYING DOCUMENTS
	417-5008	OFF (Power: disconnection from the mains)
	417-5007	ON (Power: connection to the mains)
		Warning, HIGH VOLTAGE
		Emergency Stop
		Type B

Table F-1 Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5339	X-ray Source Assembly Emitting
	417-5009	Standby
		Start
		Table Set
		Abort
		Intercom
		(on Operator Console) Power On: light on Standby: light off

Table F-1 Symbols

Appendix G

ESD Grounding Points

The VCT gantry has defined ESD attachment points that allow easy attachment for banana clip and alligator clip style connections. There are points on the stationary frame on each side of the gantry and rotating assembly behind the DAS plate. The stationary frame locations should always be used if the length of your ESD strap is sufficient.

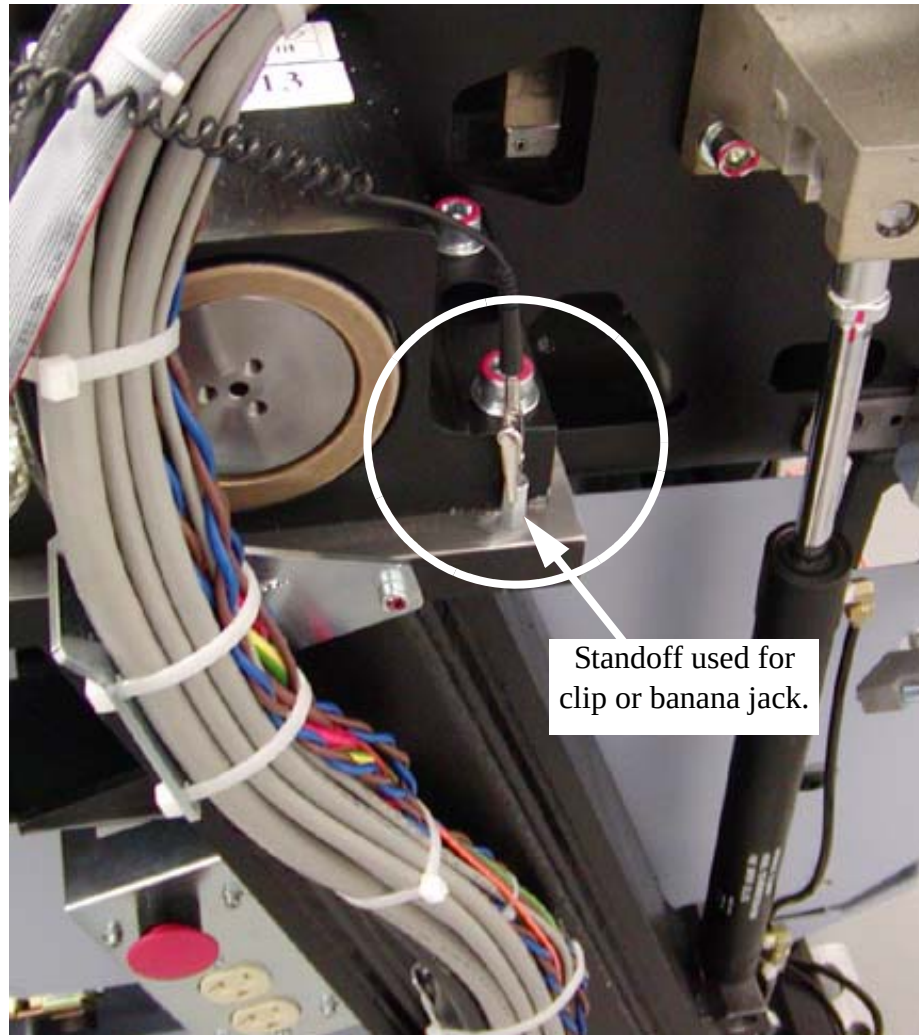


Figure G-1 Gantry Right Side Clip Example

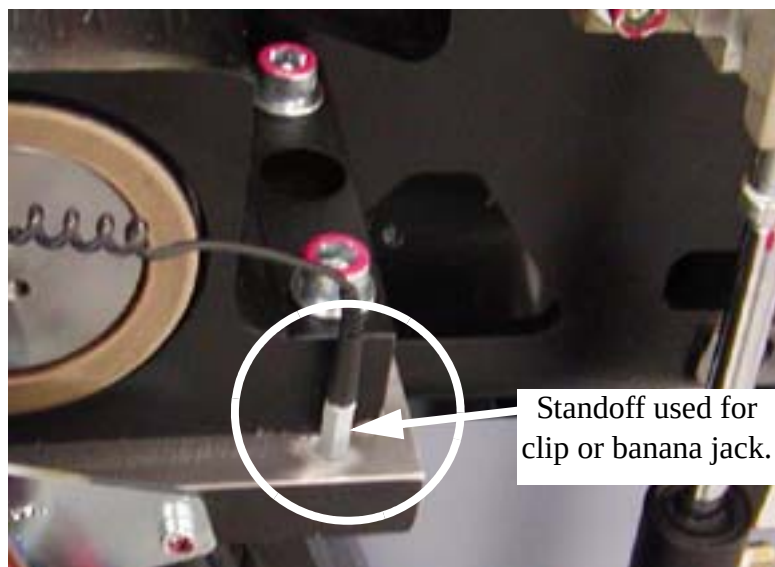


Figure G-2 Gantry Right Side Banana Jack Example

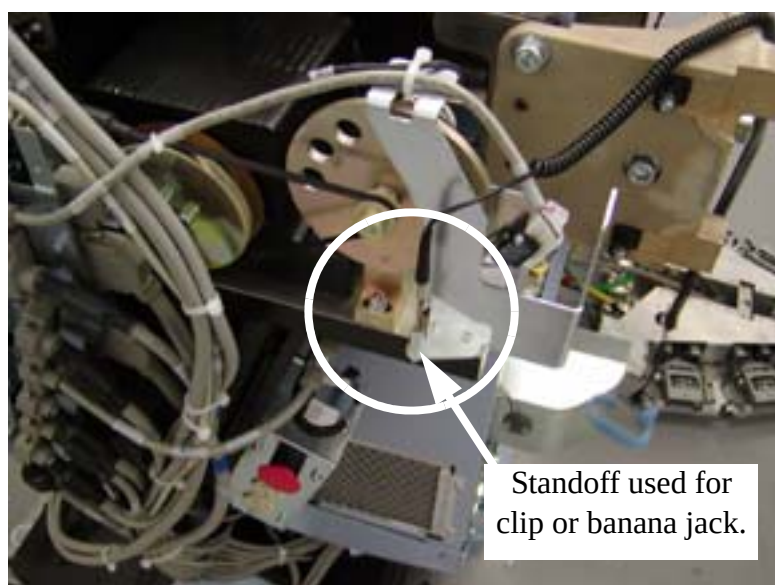


Figure G-3 Gantry Left Side Connection



Figure G-4 Rear of DAS Plate ESD Connection Points

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Appendix H

Manually Mount/Unmount a CD

Section 1.0 Introduction

This appendix contains the following procedures that may be useful during a software load:

- CDROM — Mount Process
- CDROM — Unmount Process

Section 2.0 Procedures

2.1 Preparation

- 1.) Insert a CD into the DVD ROM drive.
- 2.) Open a Shell window.
- 3.) Type: `su -` ENTER
- 4.) Type the password and press ENTER

2.2 CDROM — Manual Mount CD Process

If login was successful, continue with manually mounting the CD:

- 1.) Type: `cd /` ENTER
- 2.) Type: `mount /mnt/cdrom` ENTER
- 3.) Type: `cd /mnt/cdrom` ENTER
- 4.) The CD has now been mounted. To see what is on it, type:
`ls /mnt/cdrom` ENTER
- 5.) Type: `./copy_rpms` ENTER

The following messages appear.

```
Copying darc rpms...  
Please wait...  
Done...
```

2.3 CDROM — Manual Unmount CD Process

Unmount the CD in order to remove it from the drive; do this by typing the following:

- 1.) Type: `cd /` ENTER
- 2.) Type: `umount /mnt/cdrom` ENTER
- 3.) Type: `eject` ENTER
- 4.) Remove the CD from the DVD-ROM drive and close the drive.

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